2026 Fall

# CS270: Digital Image Processing

Register here to download lecture slides !

**Dr. Xavier LAGORCE**

**Email: xavier@shanghaitech.edu.cn**

**Office: SIST 1D-301B**

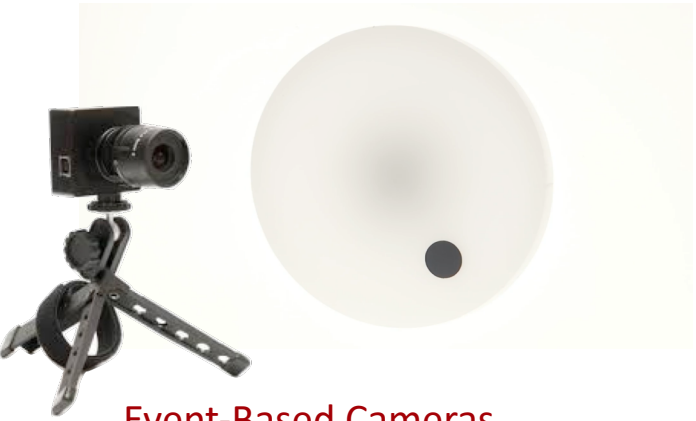
**ShanghaiTech University**

https://piazza.com/shanghaitech.edu.cn/fall2026/cs270

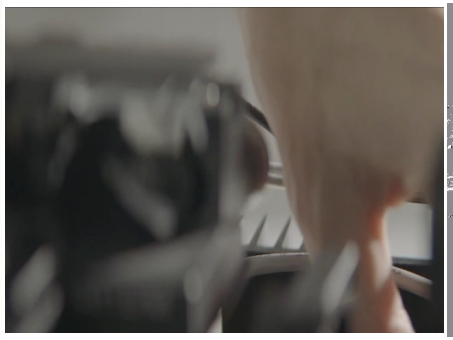
# Self introduction

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ShanghaiTech University

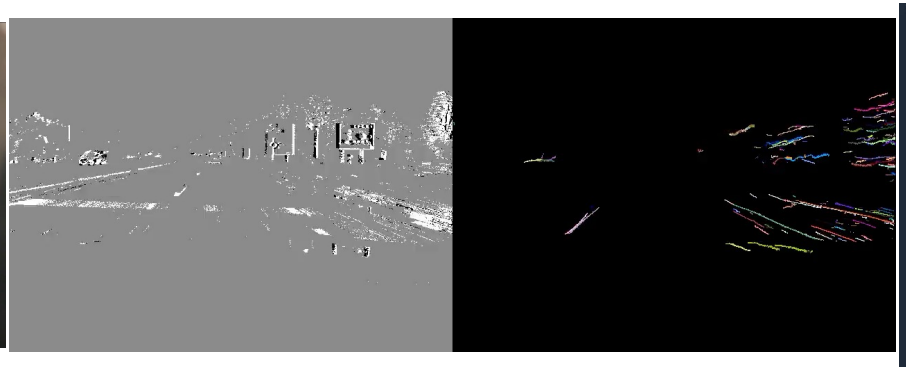
- Xavier LAGORCE, PhD
- Assistant Professor @SIST, Shanghaitech
- Research: Neuromorphic Engineering



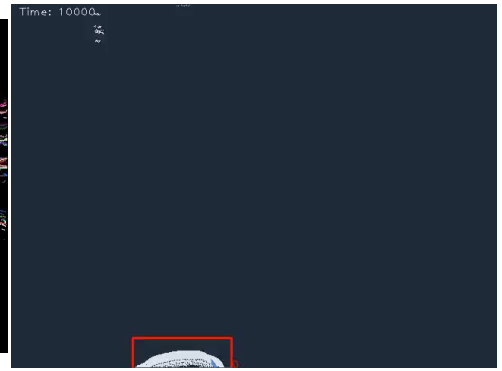
Event-Based Cameras



Active Light



Navigation



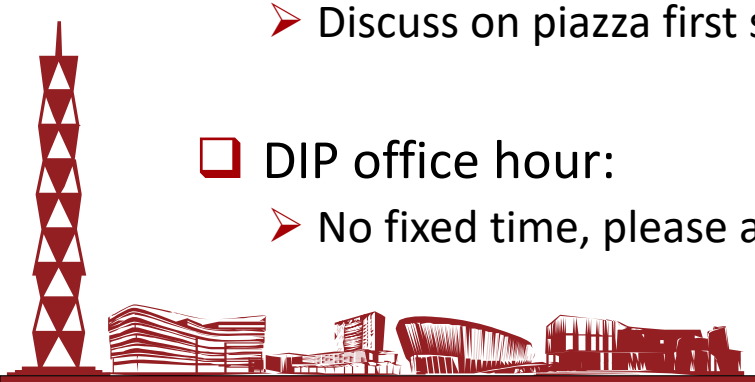

Area Monitoring


- Email: xavier@shanghaitech.edu.cn
- Office: SIST 1D-301B

# DIP support team

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- Course discussion system: Piazza
  - https://piazza.com/shanghaitech.edu.cn/fall2026/cs270
  - Please register as fast as possible, content will be shared there
- Course assignments platform: Gradescope
  - https://www.gradescope.com
  - **Details to be announced in Piazza later**
- TAs:
  - WuHao - 伍浩 - wuhao2024@shanghaitech.edu.cn
  - Probably more to come
  - Discuss on piazza first so that others can benefit too
- DIP office hour:
  - No fixed time, please ask on Piazza



# Schedule

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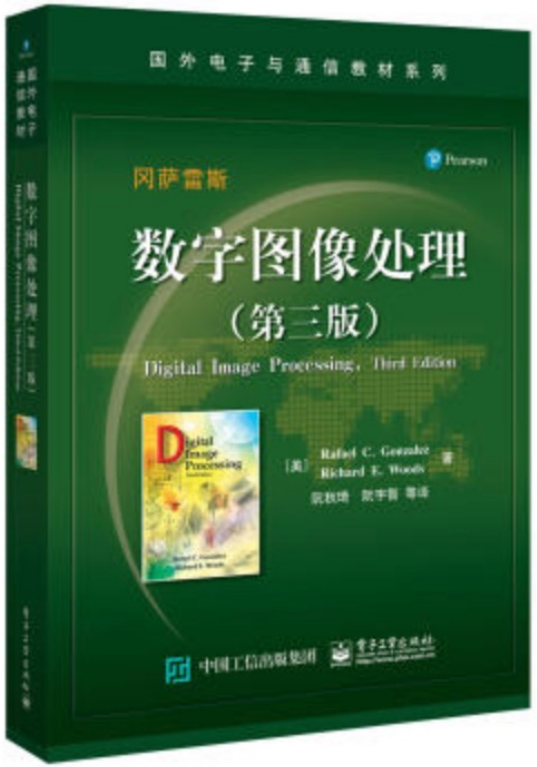
- Topic I: Digital image fundamentals (3.5w)
  - Introduction
  - Human visual system, perception and pixel
  - Intensity transform and spatial filtering
- Topic II: Transform domain processing (4.5w)
  - Frequency domain filtering
  - Image restoration and reconstruction
  - Color image processing and morphological image processing
  - Wavelet and other image transforms
- Topic III: Image segmentation (4w)
  - Edge detection, thresholding and region detection
  - Watershed and active contours
  - Feature extraction, scale invariant feature transform (SIFT)
- Course Project and Presentation (4w)



# Reference Books

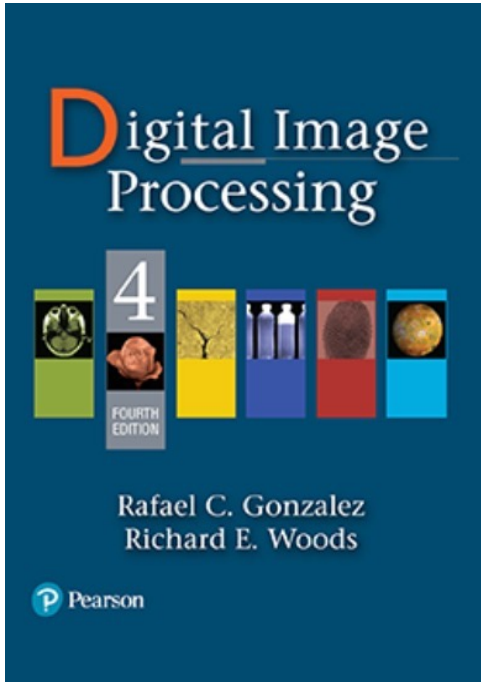
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ShanghaiTech University

- Digital Image Processing, by Rafael C. Gonzalez, & Richard E. Woods
- 978-0131687288 (3 or 4th edition)



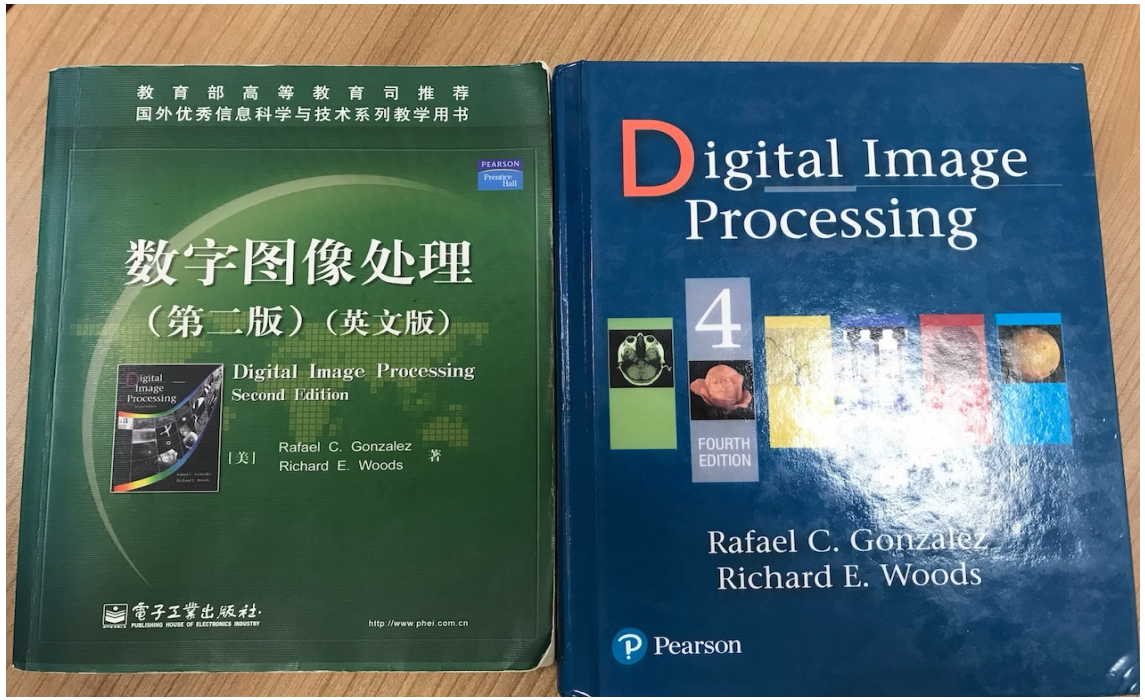


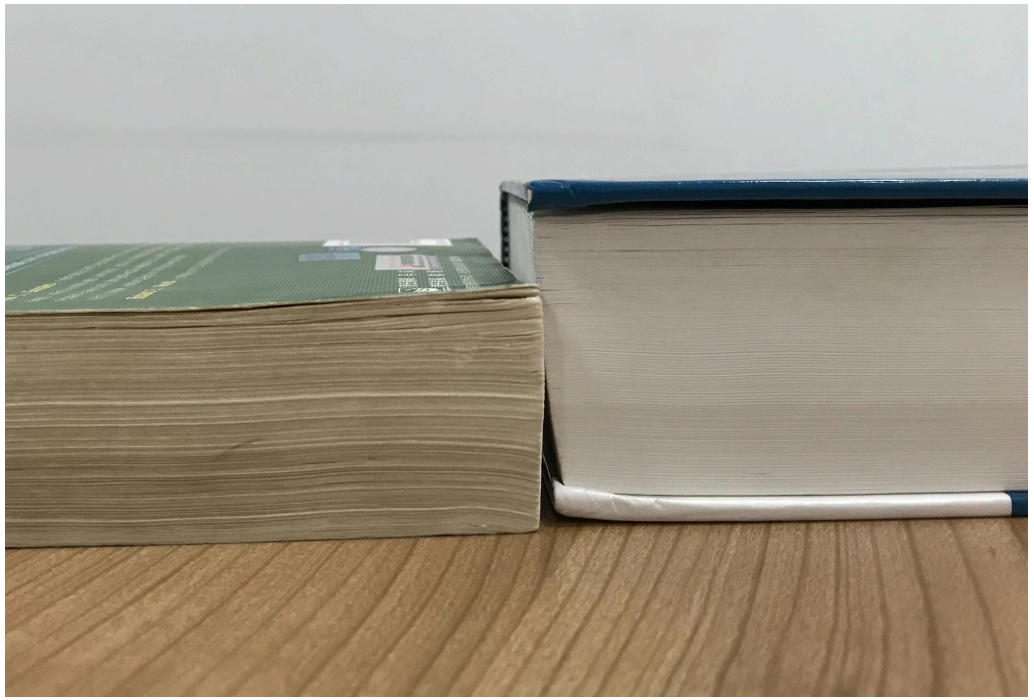




# DIP book over 10+ yrs

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Many new chapters and contents

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# Who is taking the DIP course



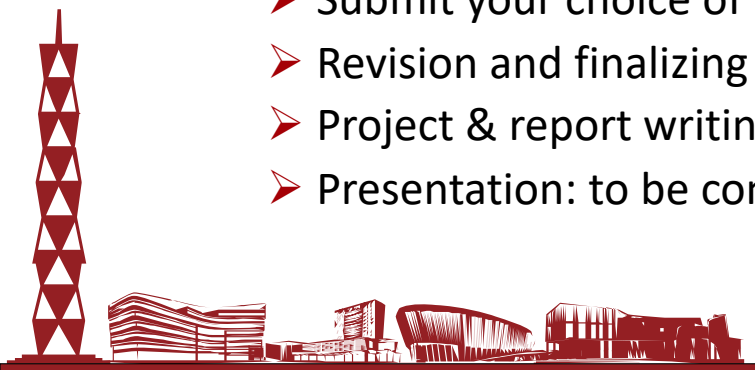
- Electronics Science and Technology: 18
- Computer Science: 16
- Electronic Information: 10
- Mechanical Engineering: 4
- Biomedical Engineering: 3
- Mathematics: 1

- Undergraduates (CS & EE): 7
- External/Visiting students: 11

# Course Assessment

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- Total (100%) = Homework (40%) + Quiz (15%) + Final Project (45%)
- Homework **(2 * 20% = 40%)**: once for first two topics; learning the theoretical basics and basic programming skills
  - **No late homework accepted!**
- Quiz **(3 * 5% = 15%)**: once for each topic; testing basic knowledges, and simple calculations; 35 min quiz in class
- Final Project **(45%)**: The evaluation is based on the quality of both presentation and report writing
  - Build your team (max 3 per team): before 10th week
  - Release of project topics: 10th week
  - Submit your choice of the project topics: 11th week
  - Revision and finalizing your project topic: 12th week
  - Project & report writing: 13th to 16th week
  - Presentation: to be confirmed



# NO plagiarism !

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- 单次抄袭：抄袭与被抄袭者该次课程任务（Homework，Quiz，Project）计零分，课程总成绩打八折。

- 累计两次抄袭：抄袭与被抄袭者课程总成绩计零，同时上报信息学院学术委员会公开处理。

- Single plagiarism: Plagiarism and the plagiarized person's course task (Homework, Quiz, Project) will be counted as zero points, and the total course score will be discounted by 20%.

- Cumulative two plagiarisms: The total score of the plagiarized and plagiarized courses will be counted as zero, and at the same time, it will be reported to the Academic Committee of the School of Information for public processing.



# The Elephant in the room…





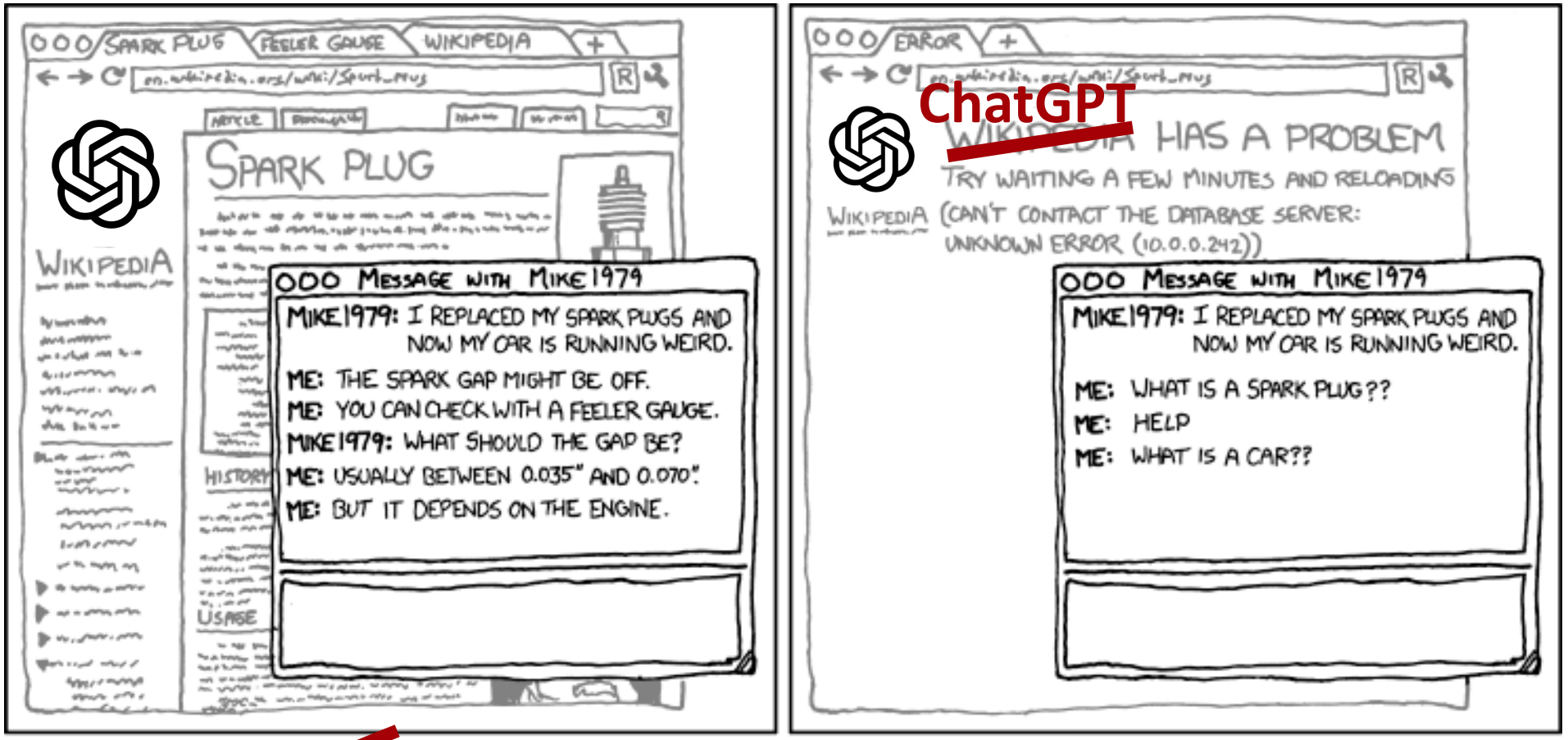
A literal elephant in a room, Sydney, 1939

AI

ChatGPT, DeepSeek, …

# The Elephant in the room…

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WHEN WIKIPEDIA HAS A SERVER OUTAGE, MY APPARENT IQ DROPS BY ABOUT 30 POINTS.

OpenAI

Adapted from https://xkcd.com/903

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# Lecture 1
# Introduction

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ShanghaiTech University

https://piazza.com/shanghaitech.edu.cn/fall2026/cs270

2026 Fall – Sept 16th

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# Image examples

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FIGURE 1.1 A digital picture produced in 1921 from a coded tape by a telegraph printer with special typefaces. (McFarlane.) [References in the bibliography at the end of the book are listed in alphabetical order by authors' last names.]



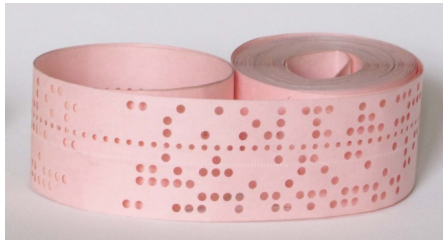
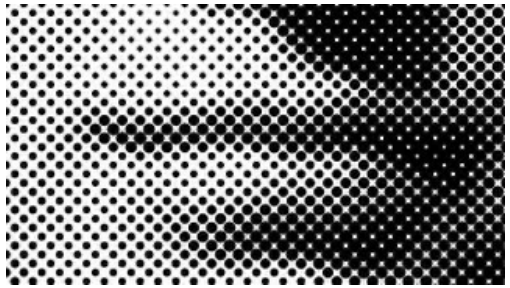




FIGURE 1.2
A digital picture made in 1922 from a tape punched after the signals had crossed the Atlantic twice. (McFarlane.)



FIGURE 1.3
Unretouched cable picture of Generals Pershing (right) and Foch, transmitted in 1929 from London to New York by 15-tone equipment. (McFarlane.)



Halftone pattern

# Image examples

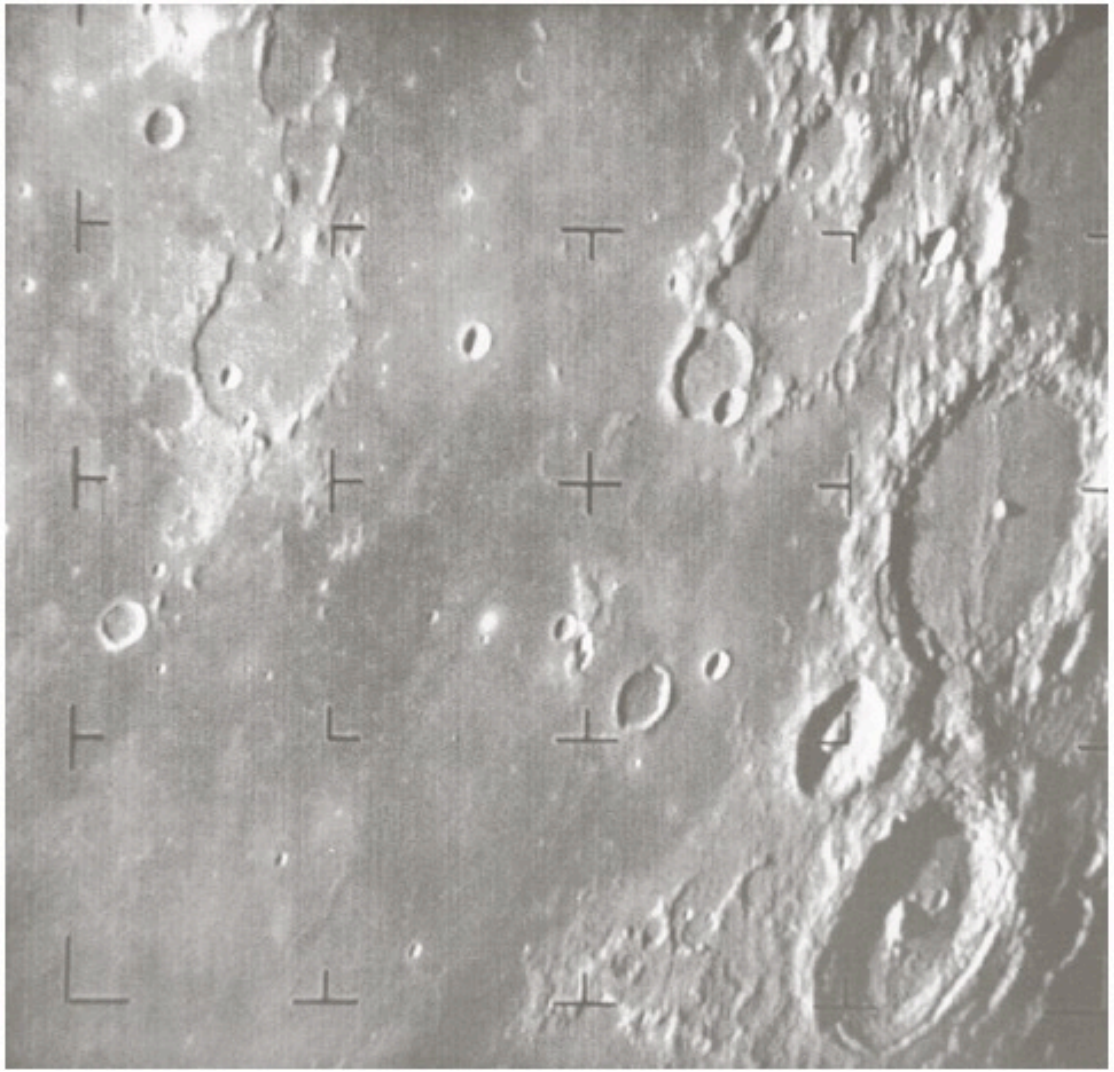


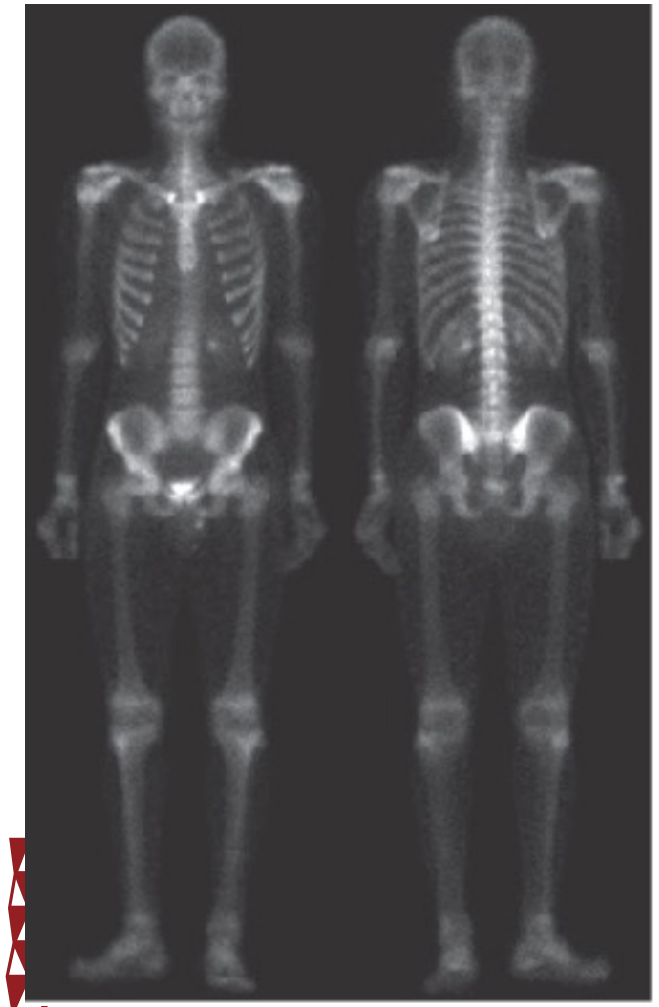
**FIGURE 1.4**
The first picture of the moon by a U.S. spacecraft. *Ranger 7* took this image on July 31, 1964 at 9:09 A.M. EDT, about 17 minutes before impacting the lunar surface. (Courtesy of NASA.)



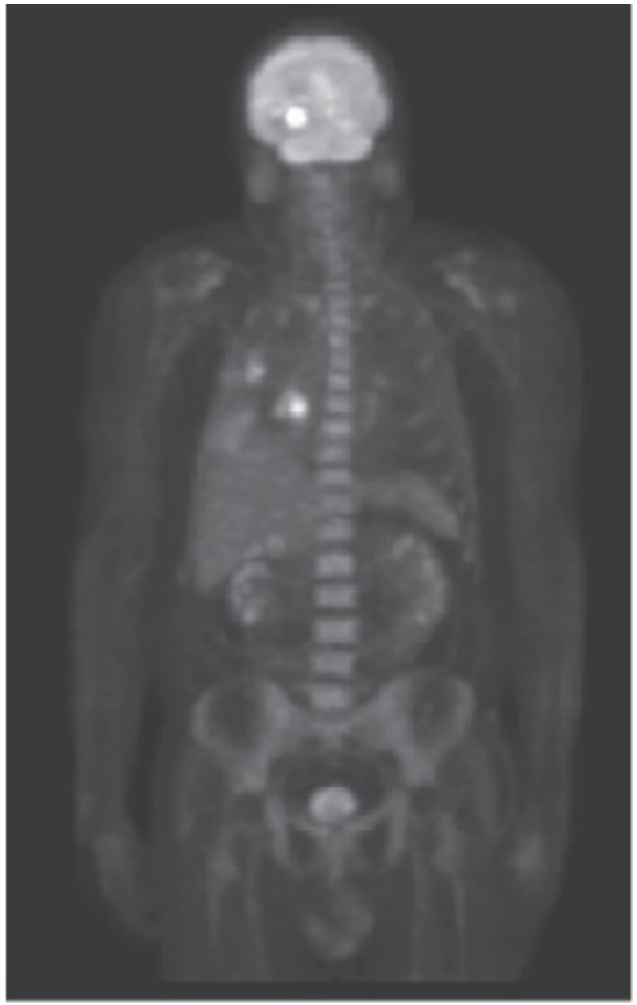
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# Image examples



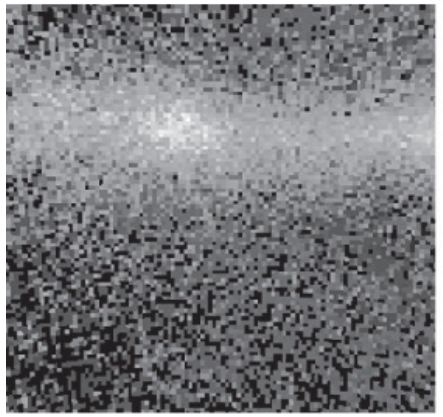


(a)

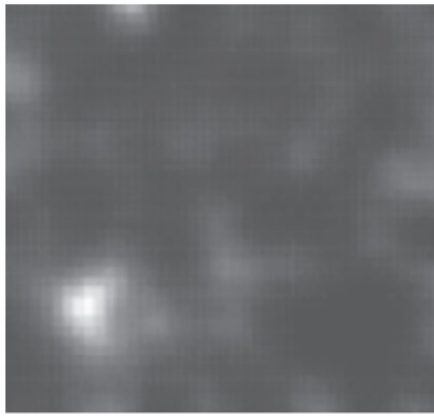


(b)

**FIGURE 1.6** Examples of gamma-ray imaging. (a) Bone scan. (b) PET image. (c) Cygnus Loop. (d) Gamma radiation (bright spot) from a reactor valve. (Images courtesy of (a) G.E. Medical Systems; (b) Dr. Michael E. Casey, CTI PET Systems; (c) NASA; (d) Professors Zhong He and David K. Wehe, University of Michigan.)



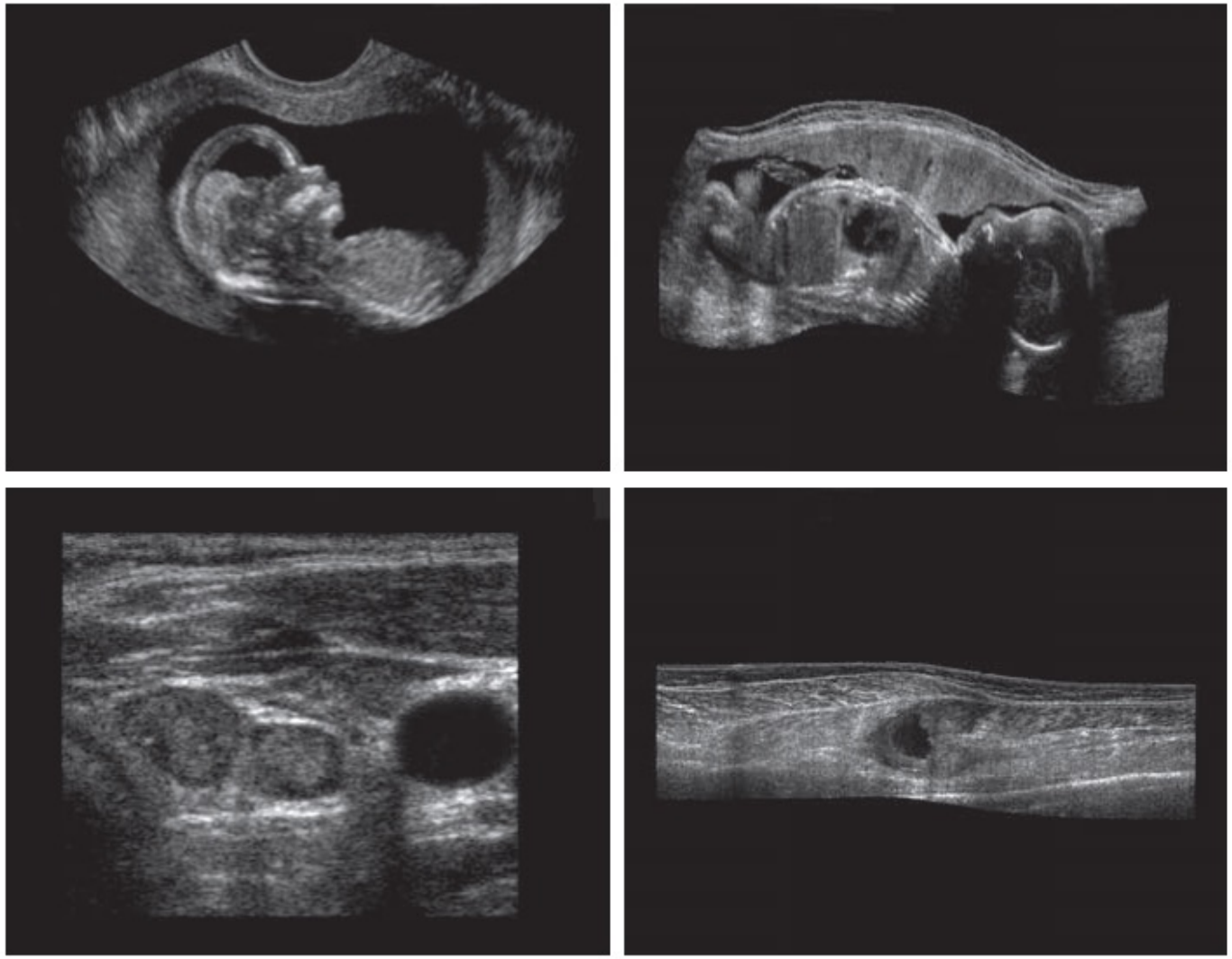
(c)



(d)

# Image examples





a b
c d

**FIGURE 1.20**
Examples of ultrasound imaging. (a) A fetus. (b) Another view of the fetus. (c) Thyroids. (d) Muscle layers showing lesion. (Courtesy of Siemens Medical Systems, Inc., Ultrasound Group.)

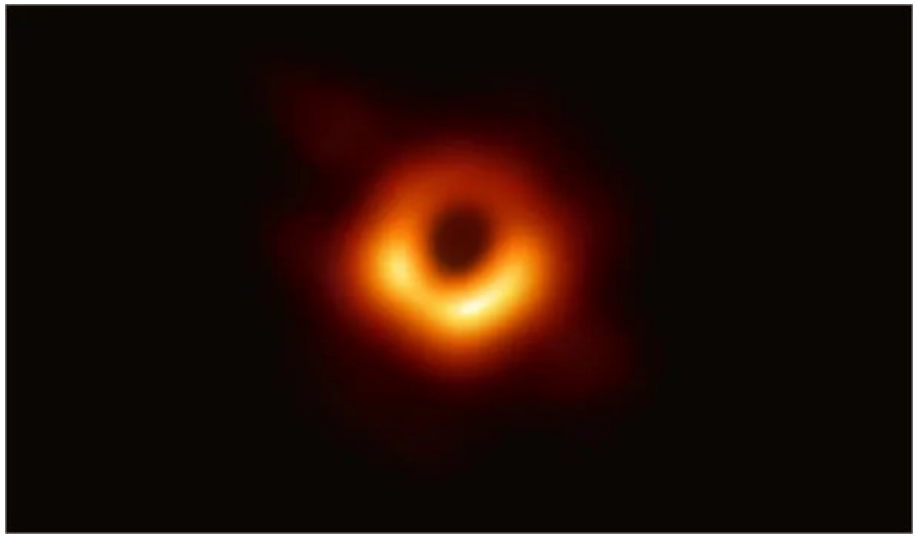


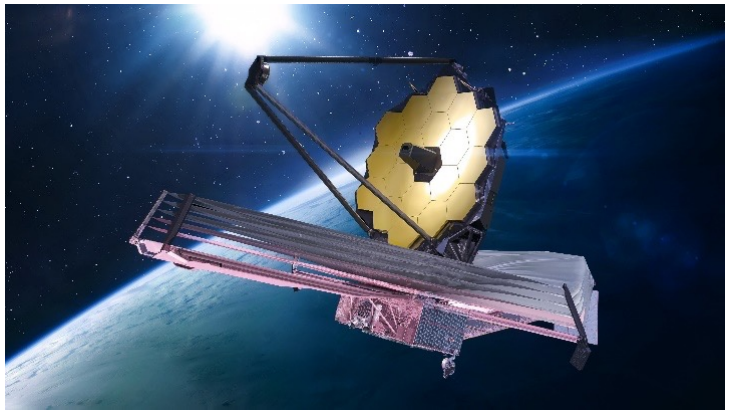


# Image examples

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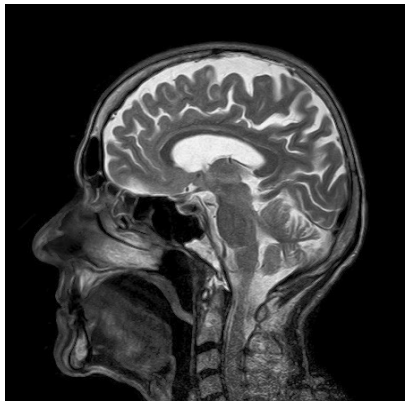
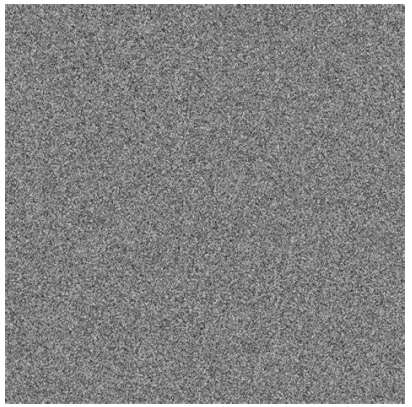


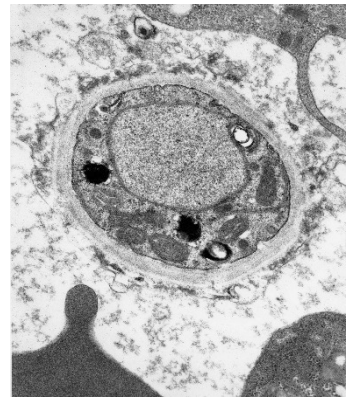
James Webb Space Telescope

# What is a Digital Image?

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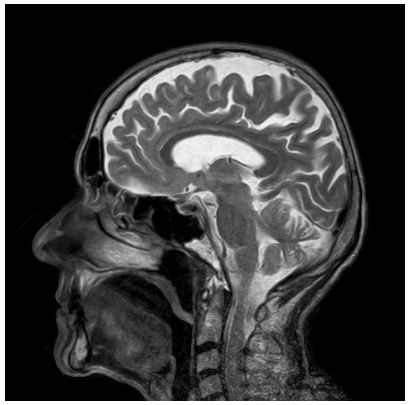
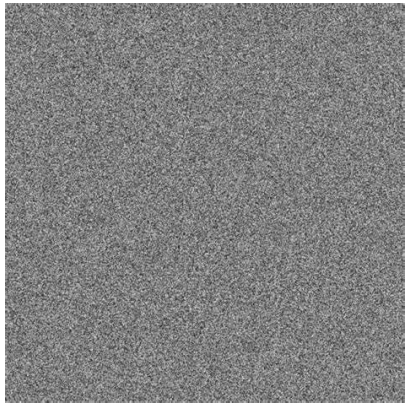
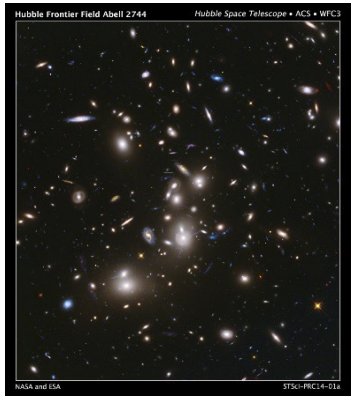



- ___?__ image
- Digital image

# What is a Digital Image?

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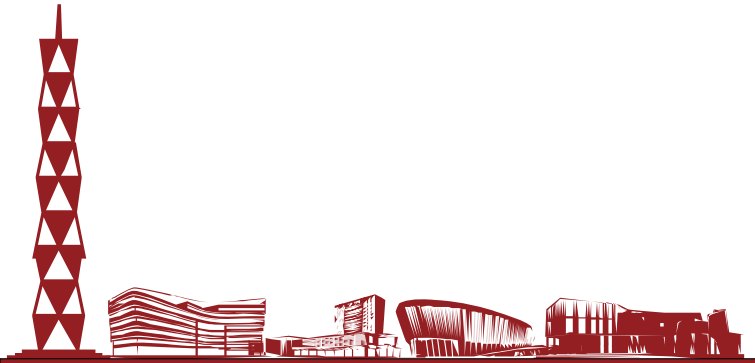











- Analog image
- Digital image

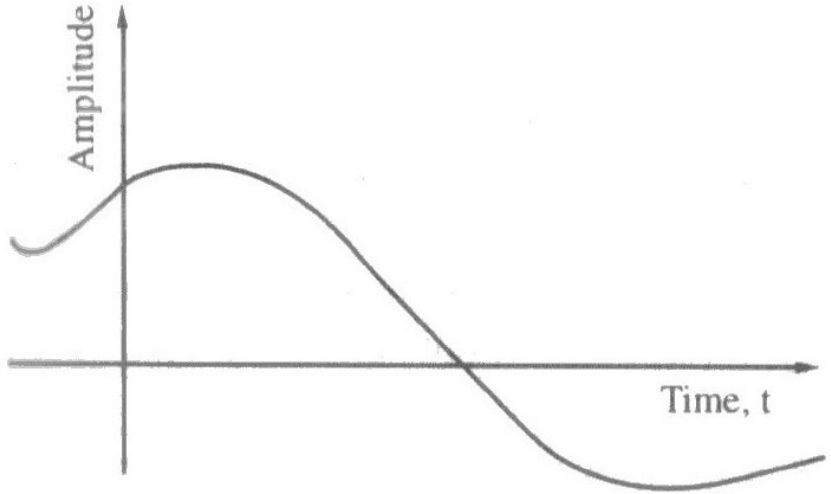


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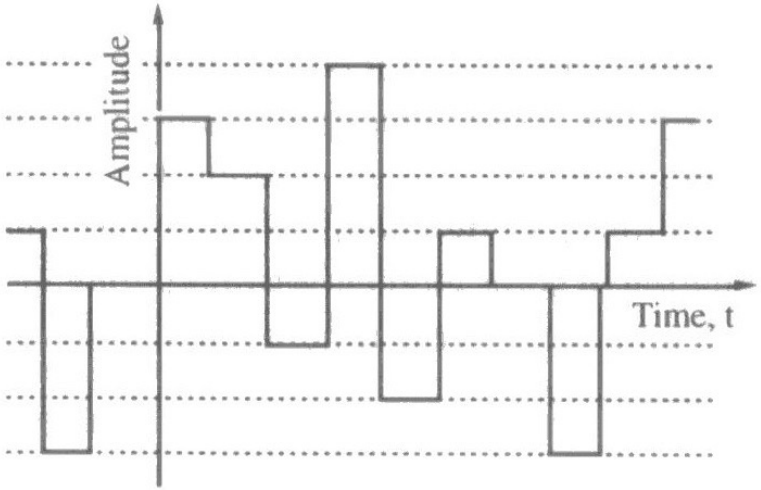
# From DSP: Analog Signal



- Analog signal
  - Continuous-time signal with continuous-valued amplitude
  - Most of the natural signals are analog



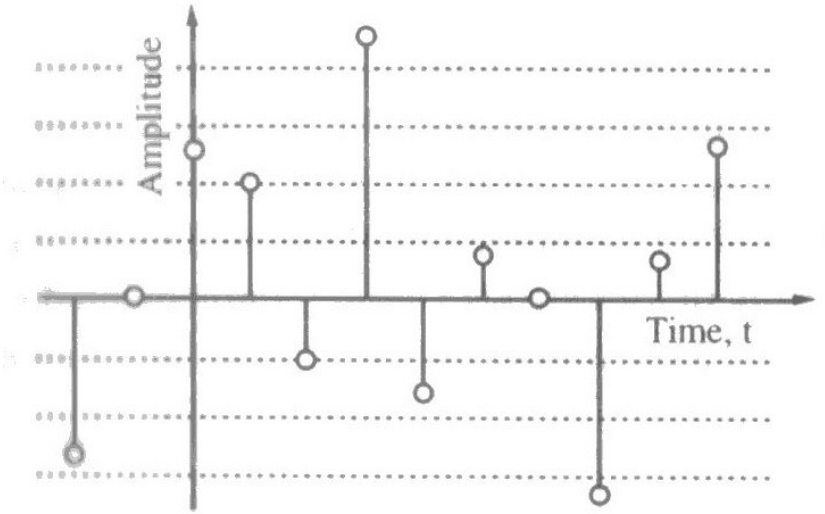

- Quantized boxcar signal
  - Continuous-time signal with discrete-valued amplitude
  - Occurs in digital electronic circuits



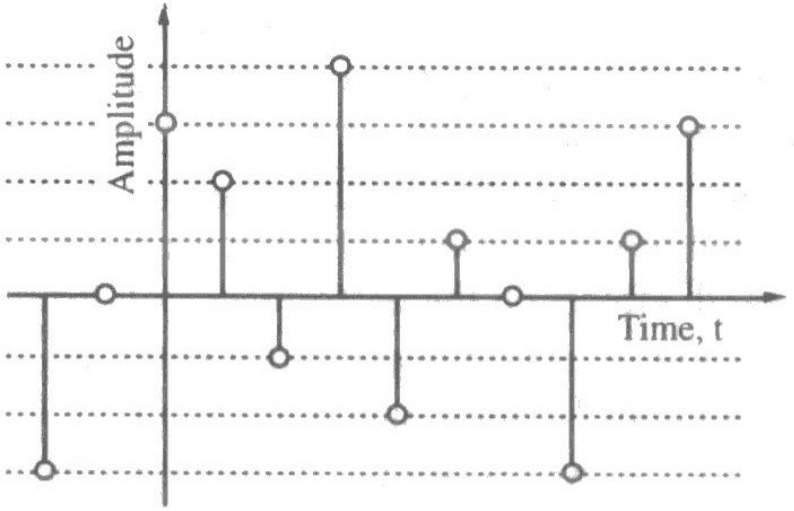
# From DSP: Digital Signal



- Sampled-data signal
  - Discrete-time signal with continuous-valued amplitude
  - The amplitude of the signal may be any value

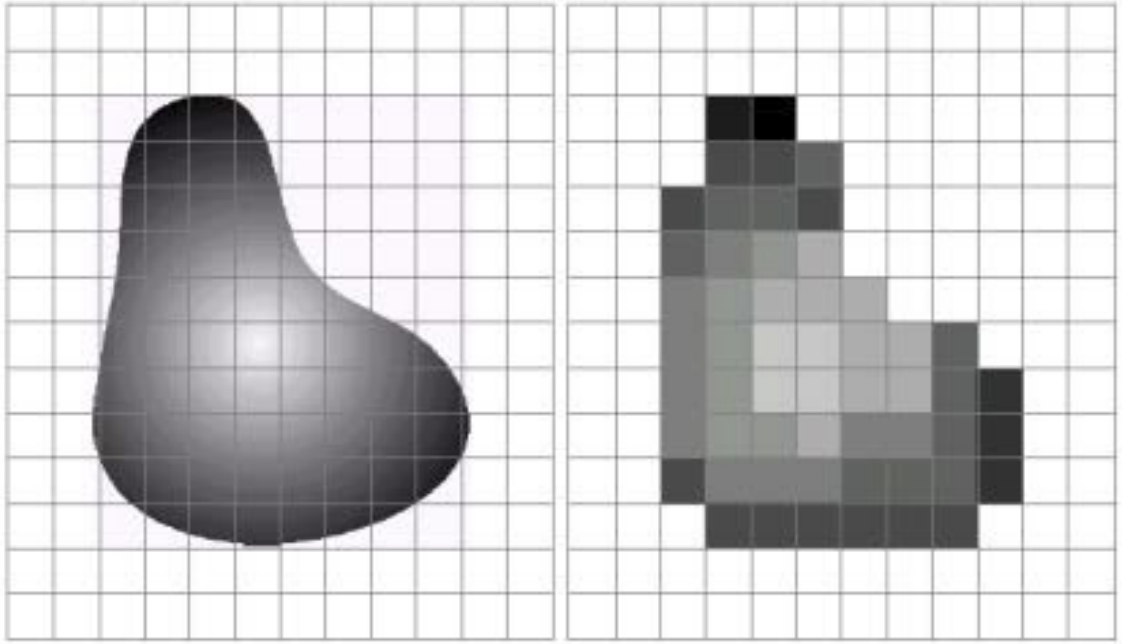



- Digital signal
  - Discrete-time signal with discrete-values amplitude
  - A digital signal is a quantized sampled-data signal

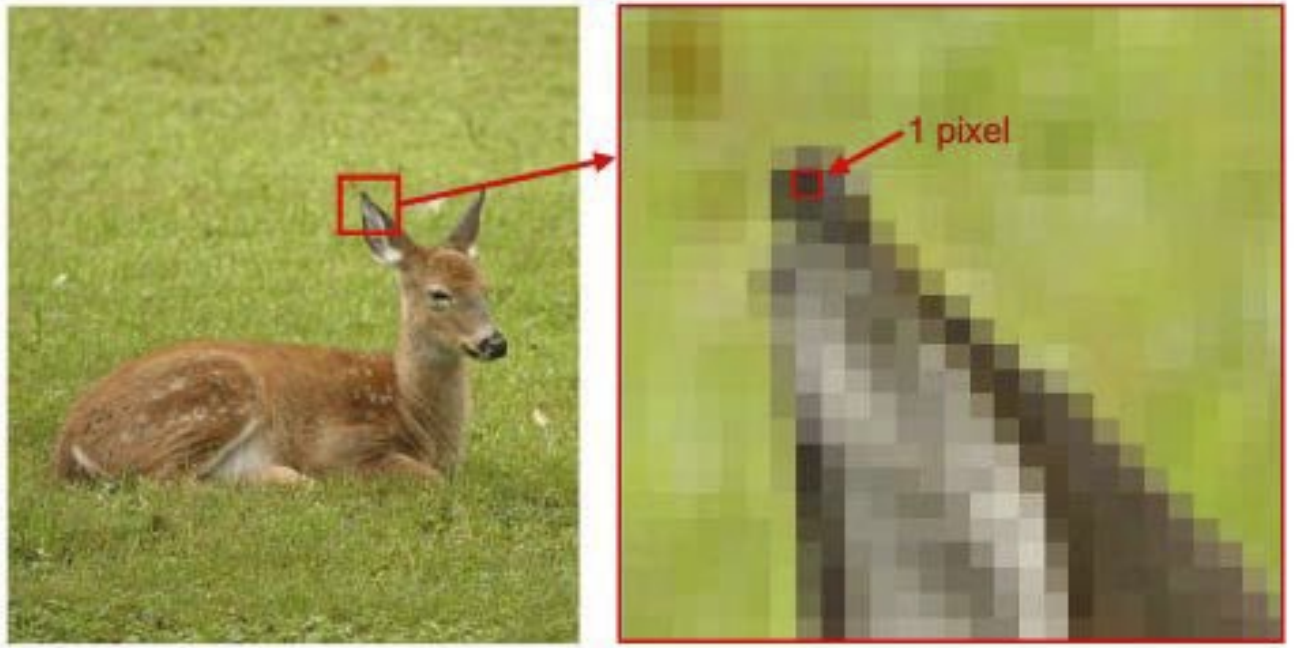


# Analog vs. Digital Image





Pixels



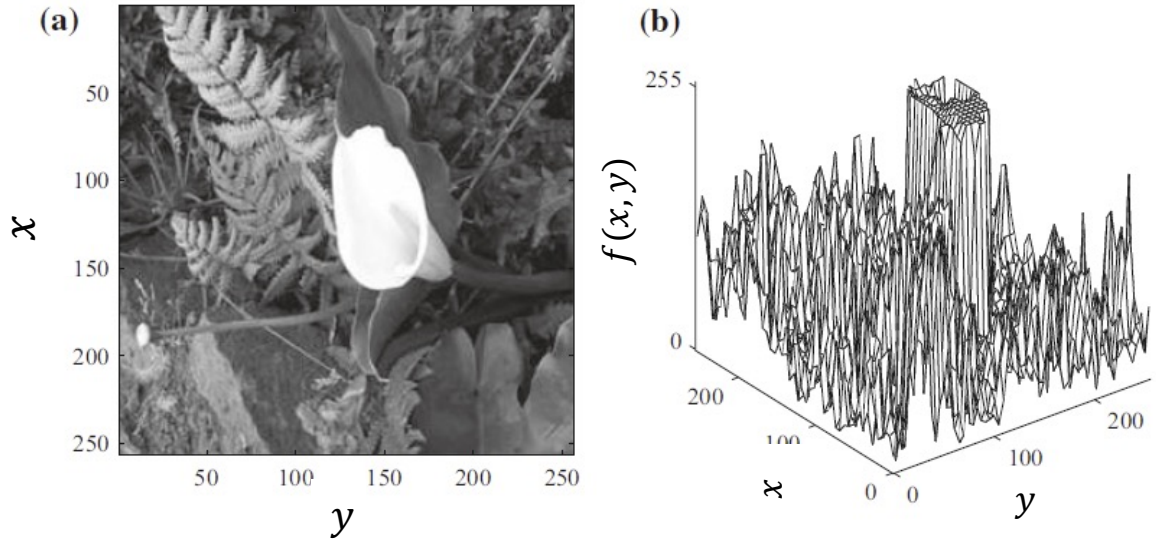

Pixels

# Pixel / Voxel

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- Digitization implies that a digital image is an approximation of a real scene.
  - Digital image composed of a finite number of elements – Pixels.
  - A pixel has a location and intensity information. It typically represents gray levels, colors, heights, opacities, etc.

- A visual representation in form of a function f(x,y), where
  - f is related to the intensity or brightness (color) at point
  - (x, y) are spatial coordinates
  - x, y and the amplitude of f are finite and discrete quantities

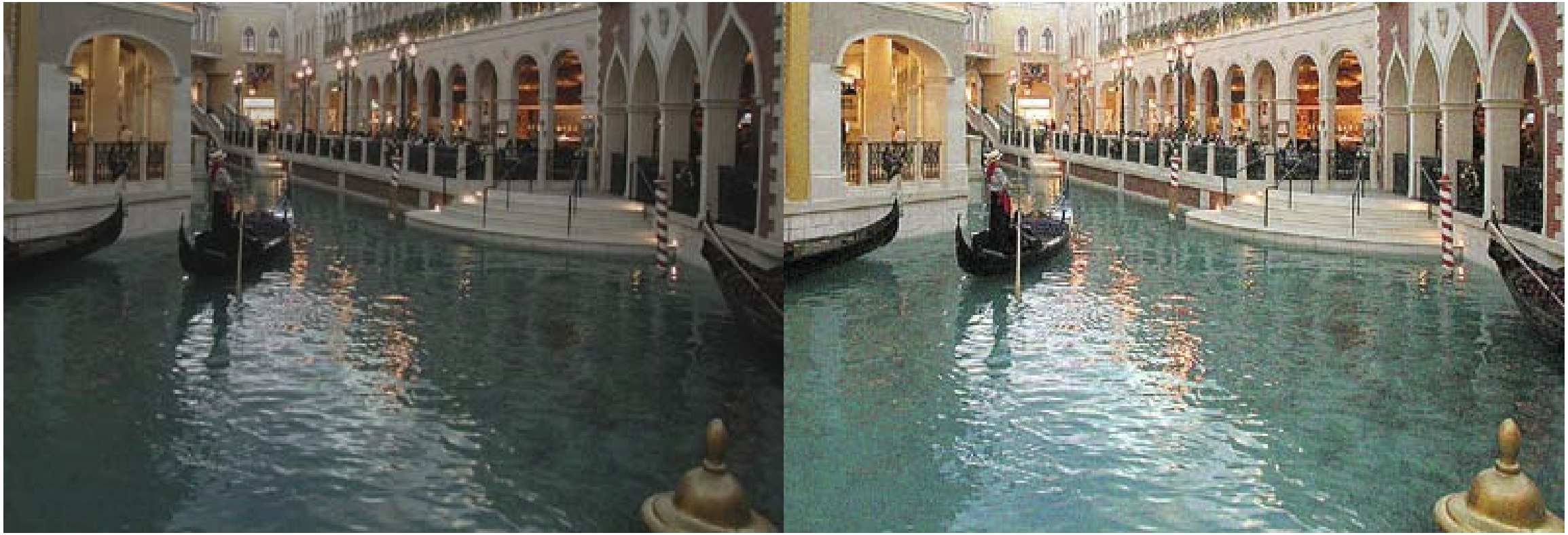



(a) A 256 × 256 image with 256 gray levels;
(b) its amplitude profile

# Digital image processing

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Definition: Processing digital images by means of a digital computer.



# Stages of DIP



| Low level process | Mid level process | High level process |
| --- | --- | --- |
| **INPUT:** Image<br>**OUTPUT:** Image<br><br>**EXAMPLE:**<br>Denoise<br>Contrast enhancement<br>Image sharpening | **INPUT:** Image<br>**OUTPUT:** Attributes<br><br>**EXAMPLE:**<br>Segmentation<br>Edge detection<br>Recognition | **INPUT:** Attributes<br>**OUTPUT:** Understanding<br><br>**EXAMPLE:**<br>Image analysis<br>Image understanding |



There are no clear-cut boundaries
from image processing to computer vision

# Stages of DIP





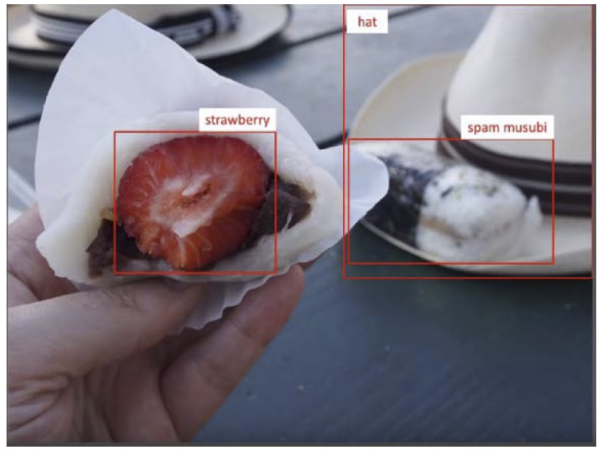
Mid level



Low level



High level



# Frequent question: relationship between Digital Image Processing and Computer Vision



| Input \ Output | Image | Knowledge |
|---|---|---|
| Image | Digital Image Processing | Computer Vision |
| Knowledge | Computer Graphic | Artificial Intelligence |

# Fundamental Steps in DIP

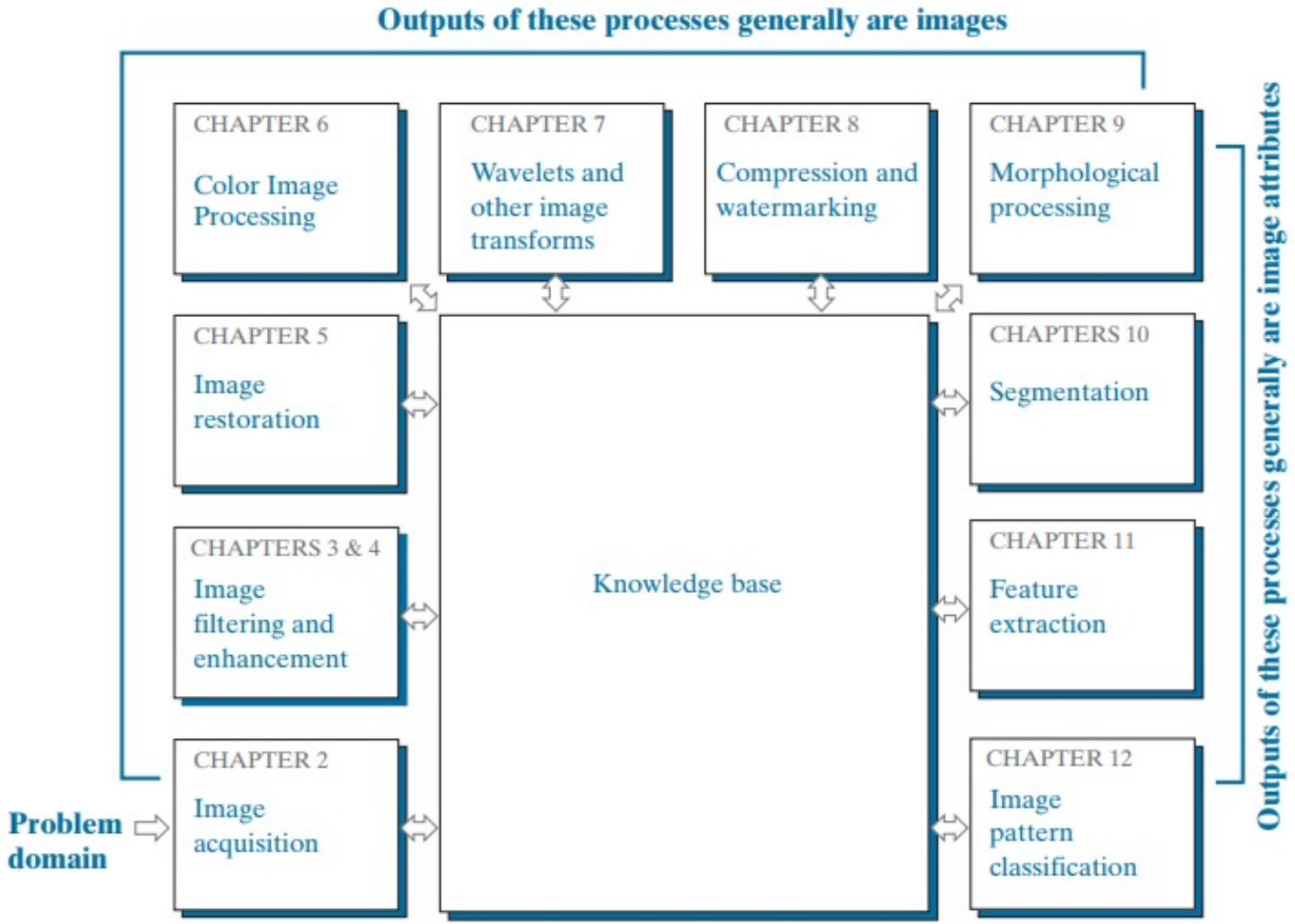


- ✓ Image acquisition
- ✓ Image reconstruction
- ✓ Image enhancement
- ✓ Image restoration
- ✓ Image compression
- ✓ Image segmentation
- ✓ Image representation and description
- ✓ Object recognition



# DIP book



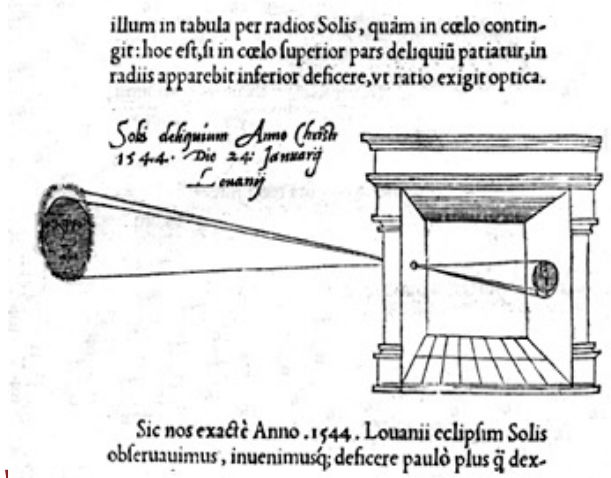

Outputs of these processes generally are images
CHAPTER 6
Color Image Processing
CHAPTER 7
Wavelets and other image transforms
CHAPTER 8
Compression and watermarking
CHAPTER 9
Morphological processing
CHAPTER 5
Image restoration
CHAPTERS 10
Segmentation
CHAPTERS 3 & 4
Image filtering and enhancement
Knowledge base
CHAPTER 11
Feature extraction
CHAPTER 2
Image acquisition
CHAPTER 12
Image pattern classification
Problem domain
Outputs of these processes generally are image attributes

# Image acquisition

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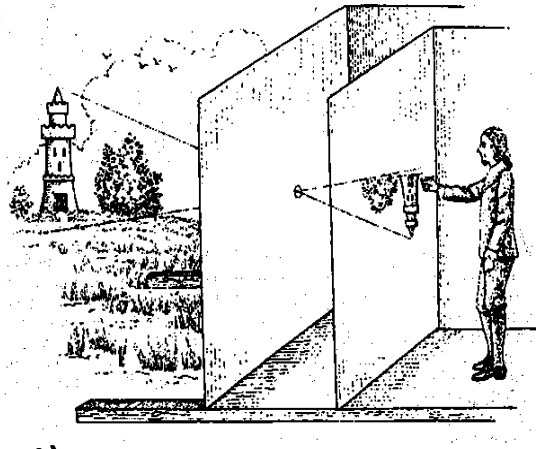



Camera Obscura

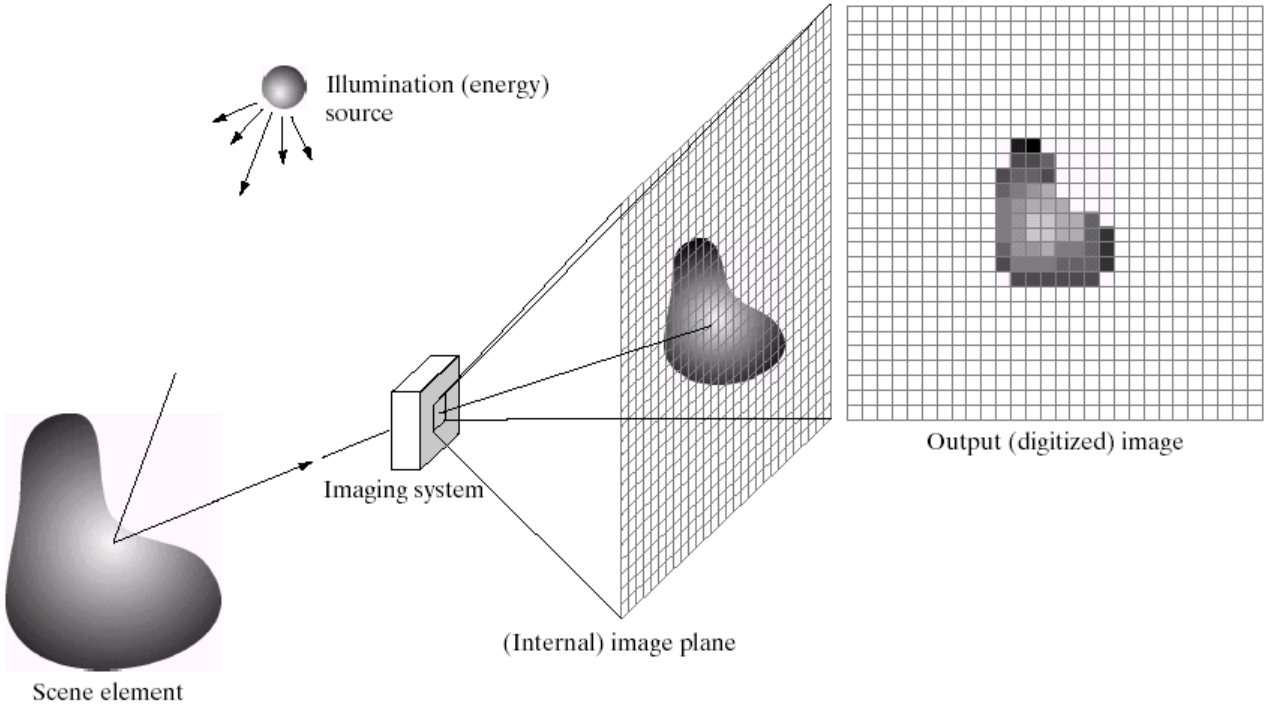

illum in tabula per radios Solis, quàm in cœlo contingit: hoc eſt, ſi in cœlo ſuperior pars deliquiũ patiatur, in radiis apparebit inferior deficere, vt ratio exigit optica.

Solis deliquium Anno Chriſti 1544. Die 24. Januarij Louanij

Sic nos exactè Anno .1544. Louanii eclipſim Solis obſeruauimus, inuenimusq; deficere paulò plus q̃ dex-



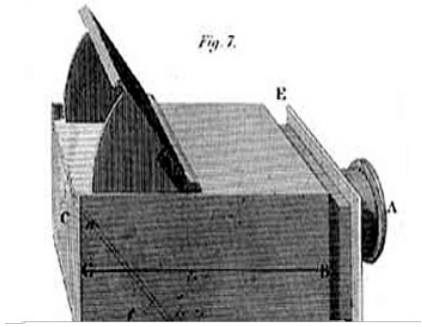
1544 (L. Da Vinci)



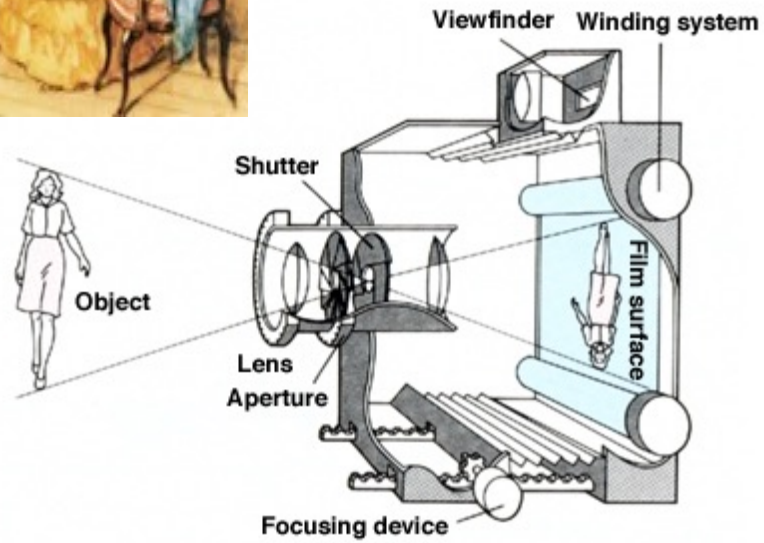
# Image acquisition









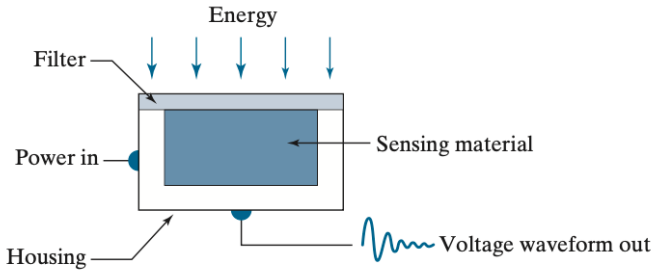



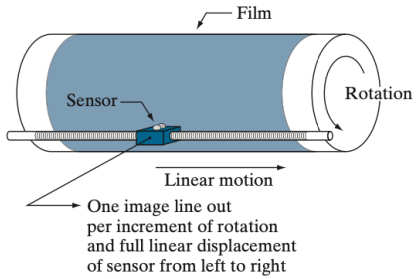




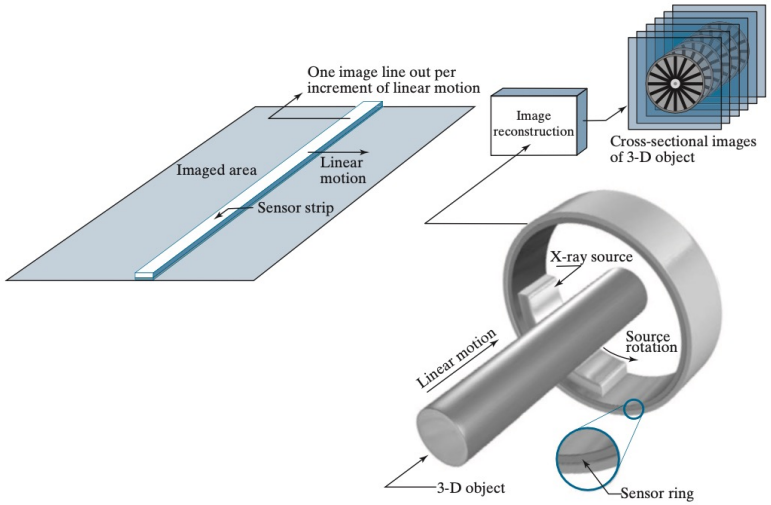

# Imaging Sensors

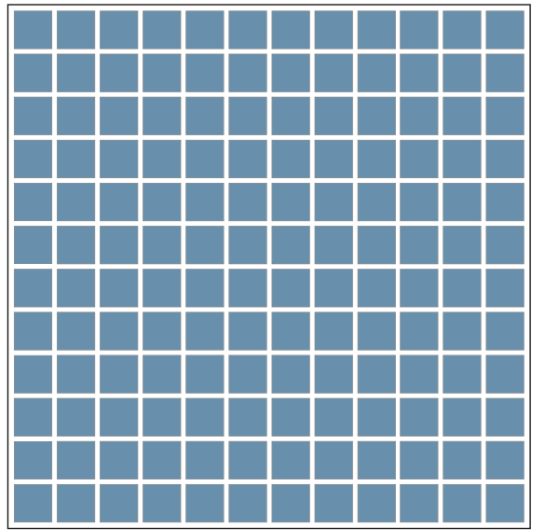






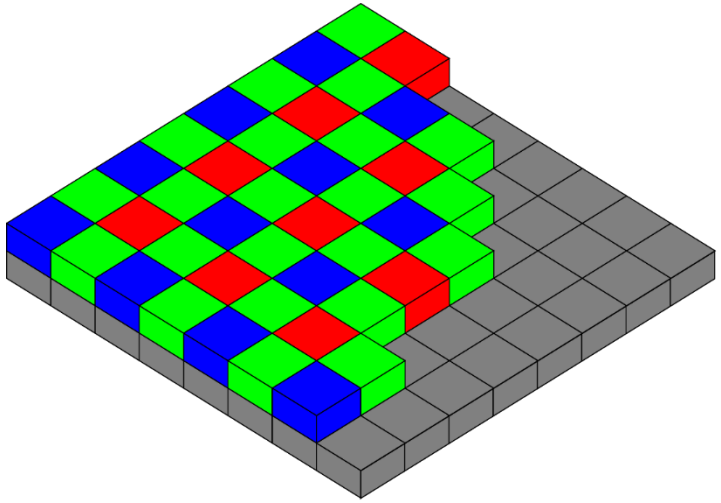



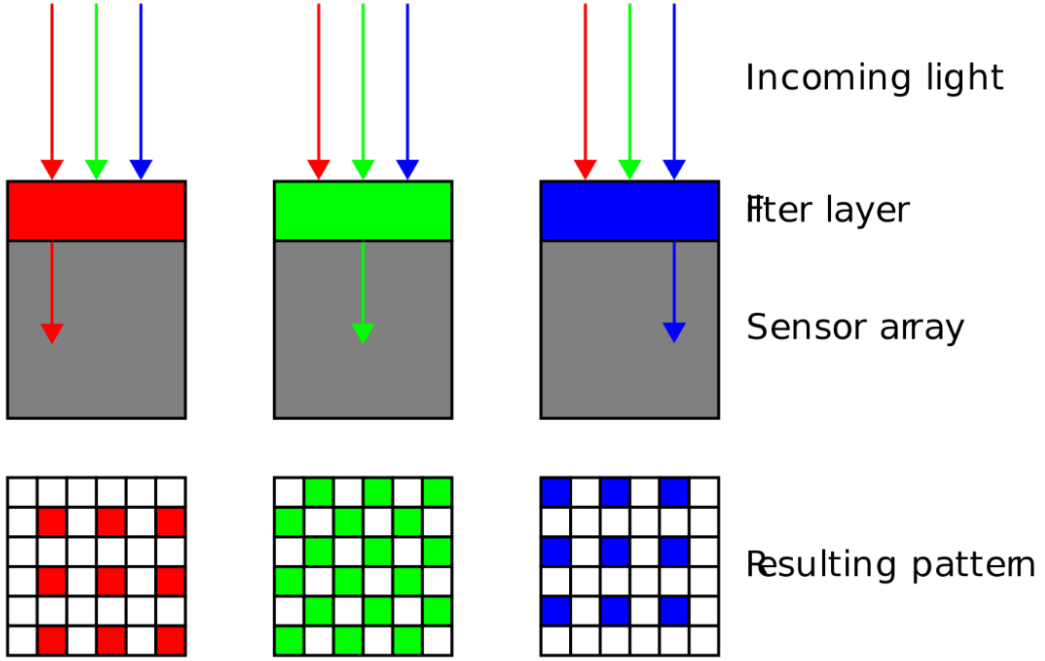
## Transform energy to voltage:

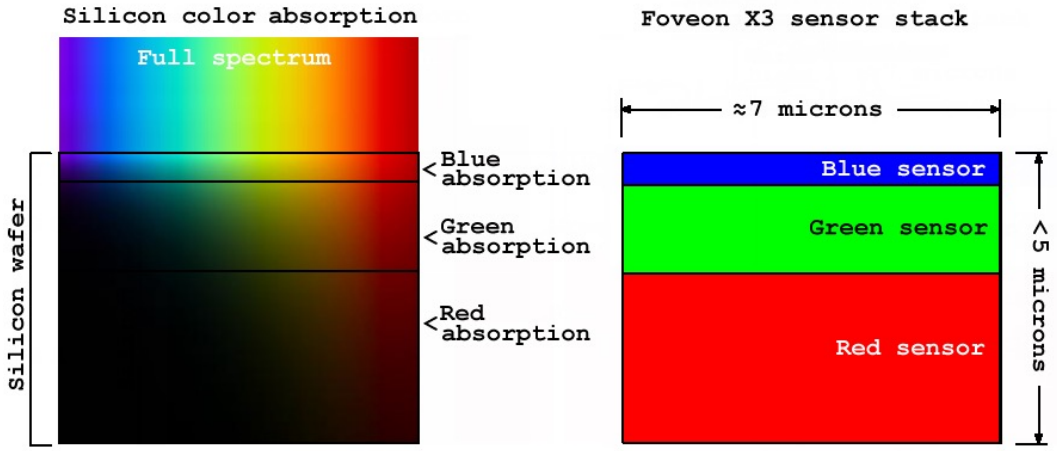
- **Single Sensor**
  - Photodiode
  - Piezoelectric element
- **Sensor Strips**
  - CAT (Computerized axial tomography)
  - Airborne imaging
  - Ultrasound array transducer
- **Sensor Array**
  - CCD (Charge-coupled Device) – digital camera
- **Color Images**
  - Use a Bayer's mosaic pattern of R/G/B filters to reduce cost, then use demosaicing to construct full resolution color images.

# Bayer filter








$$f(\cancel{x, y}) \longrightarrow f(x, y, c)$$

https://en.wikipedia.org/wiki/Foveon_X3_sensor

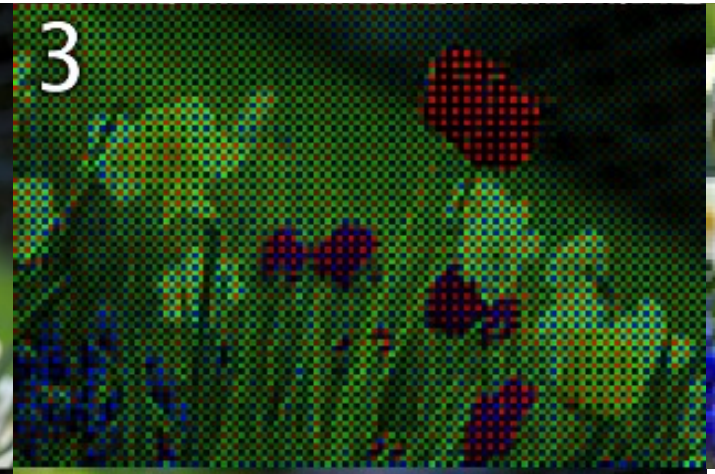
# Color coded and reconstruction







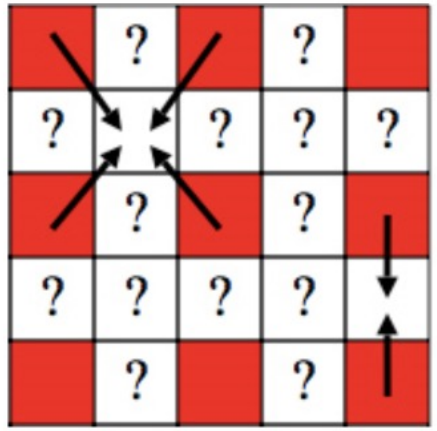


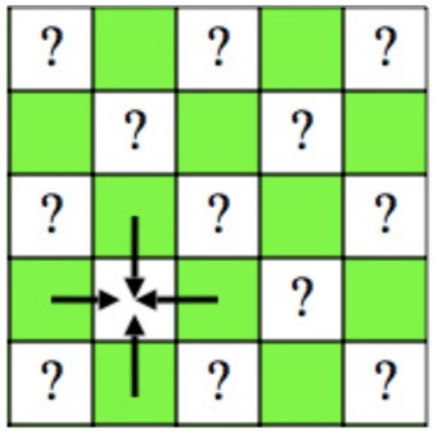










# Sensor size comparisons for digital camera

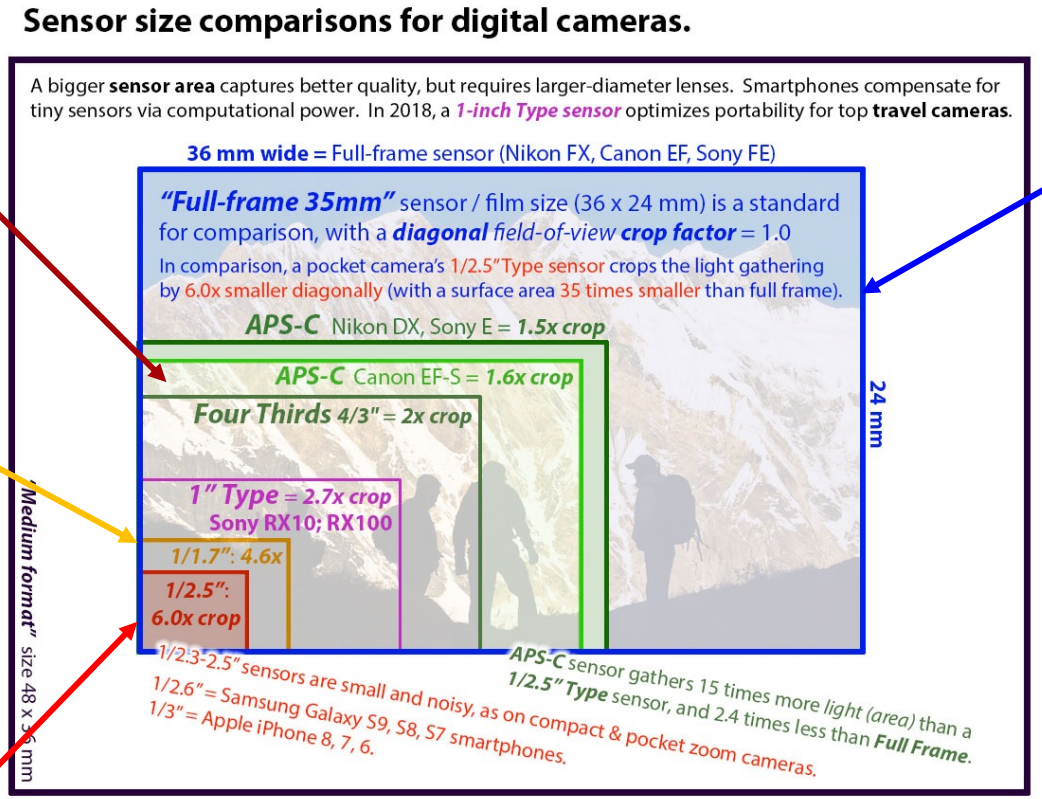












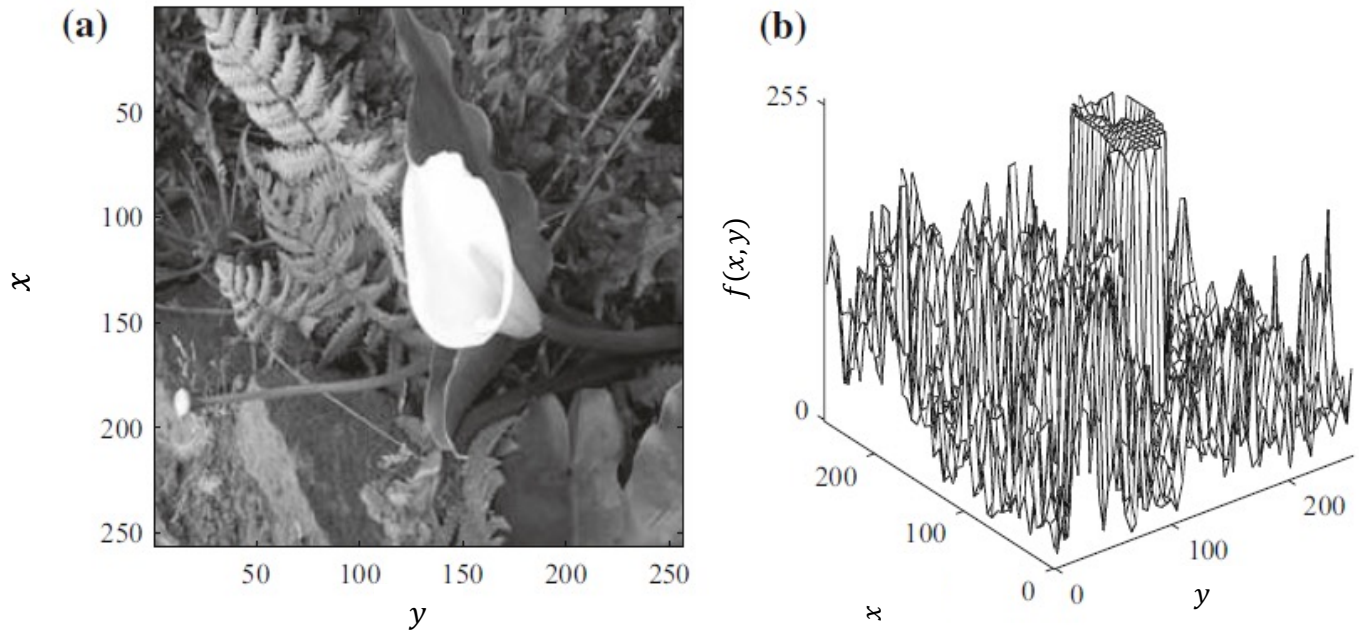




# Digital image

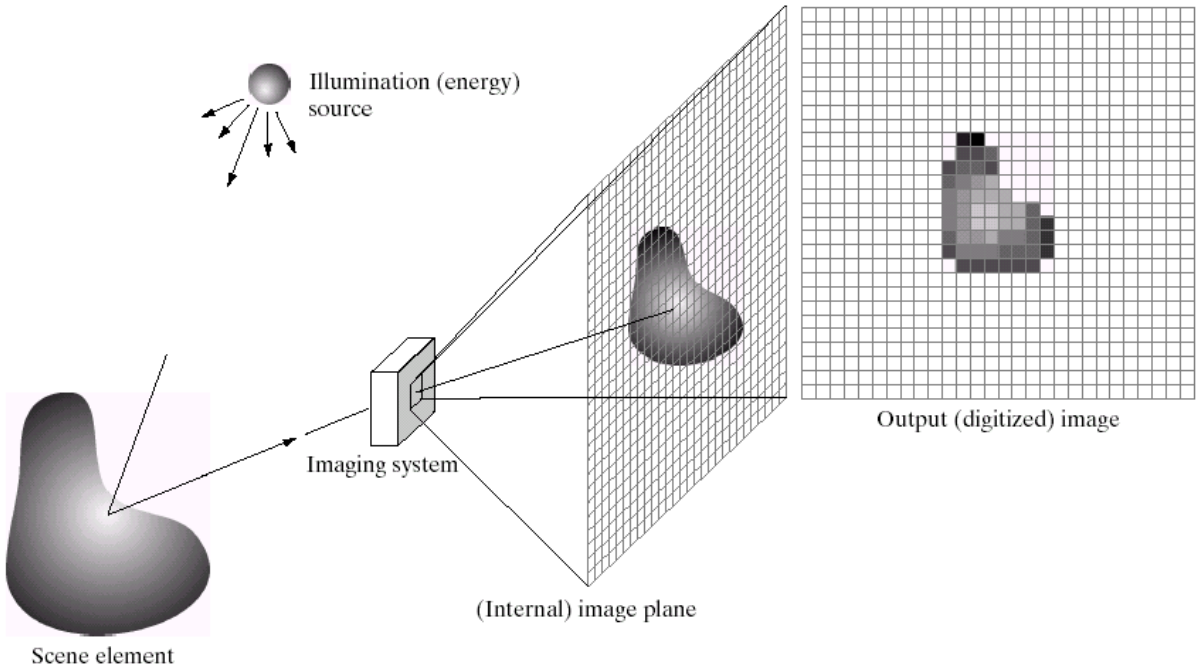
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- **A visual representation in form of a function $\mathbf{f(x, y)}$, where**
  - $f$ is related to the intensity or brightness (color) at point
  - $(x, y)$ are spatial coordinates
  - $x, y$, and the amplitude of $f$ are finite and discrete quantities




(a) A 256X256 image with 256 gray levels; (b) its amplitude profile

# Illumination model




$$f(x,y) = i(x,y)r(x,y) \qquad 0 < i(x,y) < \infty, 0 \le r(x,y) < 1$$

$$L_{min} < f(x_0,y_0) < L_{max} \qquad \text{where } L_{min} \text{ is positive}, L_{max} \text{ is finite}$$

# Illumination model



$$f(x,y) = i(x,y)r(x,y) \qquad 0 < i(x,y) < \infty, 0 \le r(x,y) < 1$$

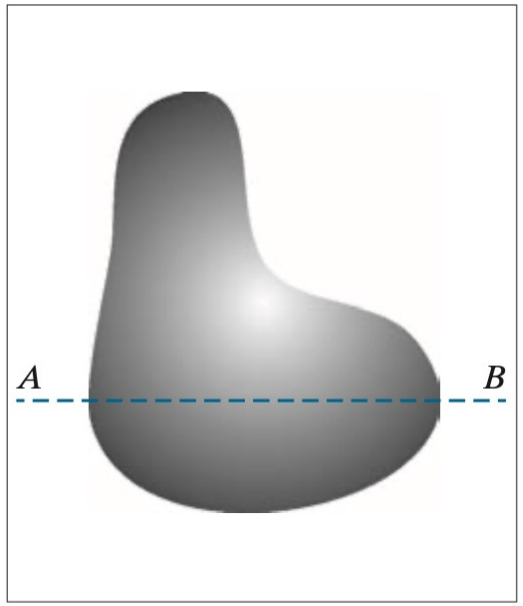
$$L_{min} < f(x_0, y_0) < L_{max} \qquad \text{where } L_{min} \text{ is positive, } L_{max} \text{ is finite}$$

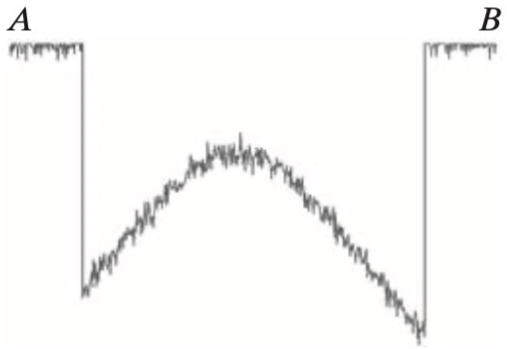
- Illuminations: $i(x,y)$ W/m2
  - Clear sunny day: 90k
  - Cloudy day: 10k
  - Indoors: 1k
  - Full Moon: 0.1
- Reflections: $r(x,y)$
  - Snow: 0.93
  - Flat white wall: 0.80
  - Stainless steel: 0.65
  - Black cloth: 0.01

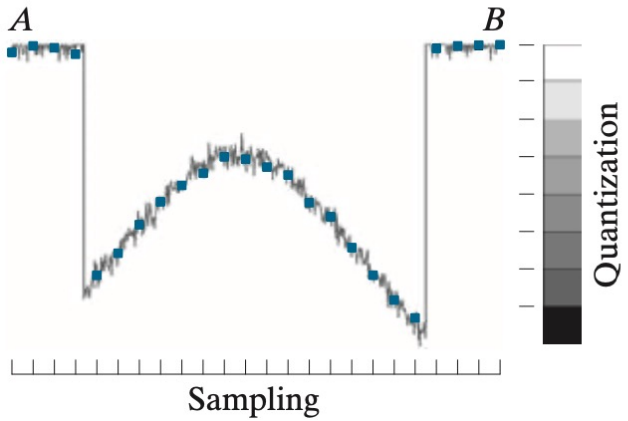


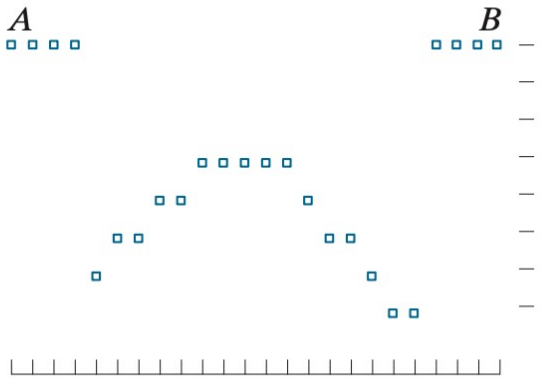
# Sampling and Quantization





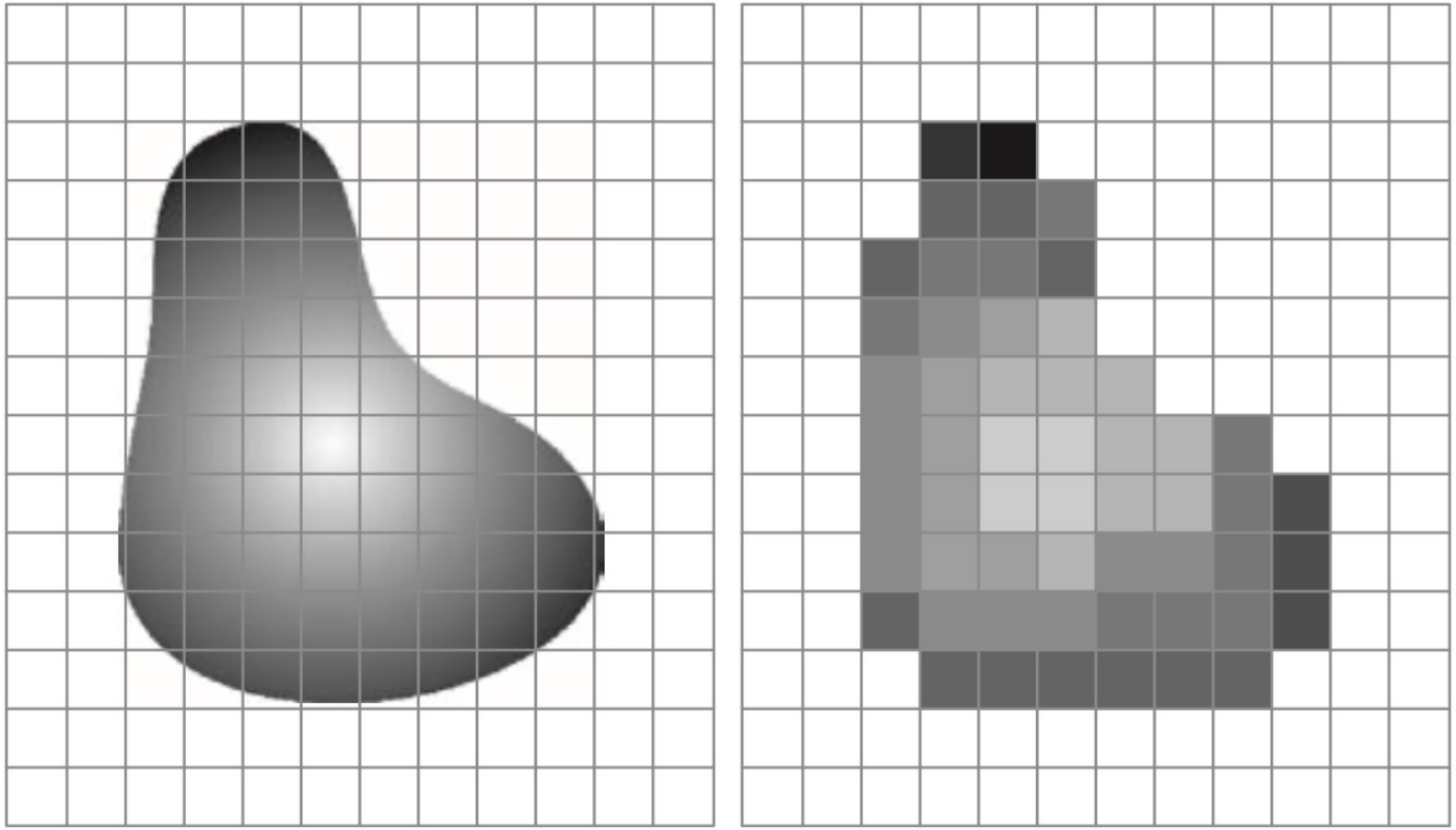




- Sampling:
  - *Digitize the coordinate values*
- Quantization:
  - *Digitize the amplitude values*
- Both can be
  - *Uniform*
  - *Non-uniform*

# Sampling and Quantization



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# Matrix Representation

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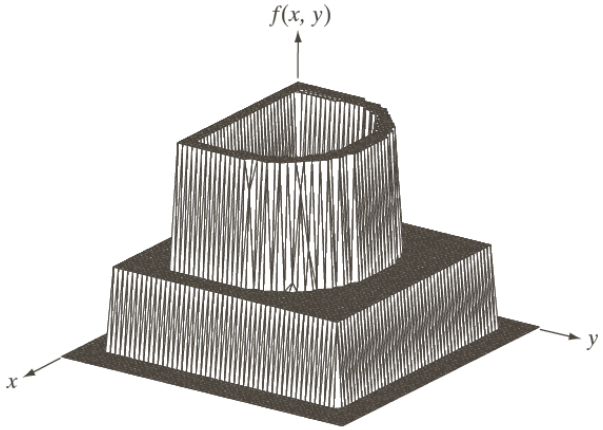
**Three basic ways to represent *f*(*x*, *y*)**

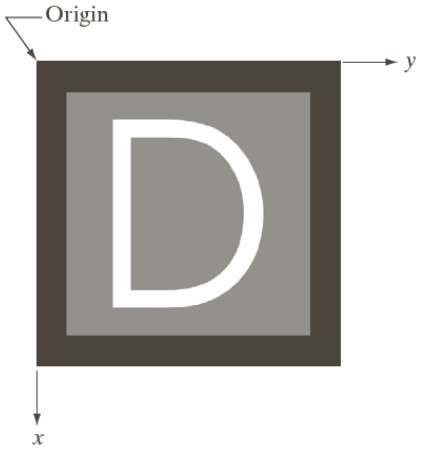
- Plot of function: *difficult to view and interpret*
- Visual intensity array: *for view*
- numerical array: *for processing and algorithm development*

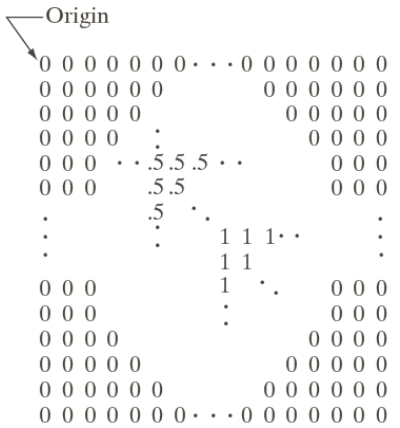
$$[f(x,y)] = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \ddots & \cdots & \vdots \\ f(M-1,0) & f(M-1,1) \cdots & \cdots & f(M-1,N-1) \end{bmatrix}$$

$$A = \begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,N-1} \\ \vdots & \ddots & \cdots & \vdots \\ a_{M-1,0} & a_{M-1,1} \cdots & \cdots & a_{M-1,N-1} \end{bmatrix}$$

Intensity level $L = 2^k$, then $b = M \times N \times k$



# Matrix Representation



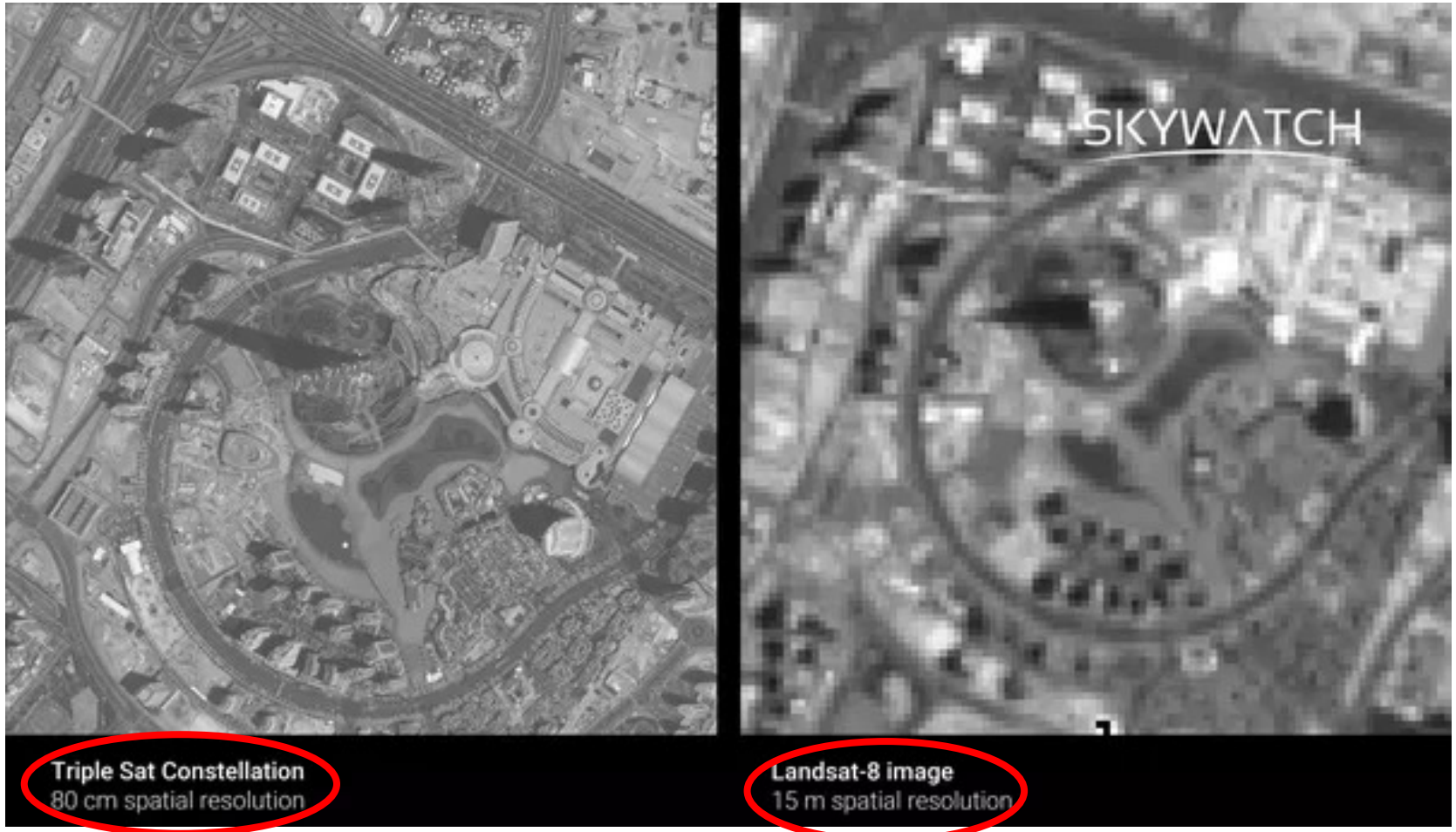
Number of storage bits for various values of $N$ and $k$.

| $N/k$ | 1 ($L = 2$) | 2 ($L = 4$) | 3 ($L = 8$) | 4 ($L = 16$) | 5 ($L = 32$) | 6 ($L = 64$) | 7 ($L = 128$) | 8 ($L = 256$) |
|---|---|---|---|---|---|---|---|---|
| 32 | 1,024 | 2,048 | 3,072 | 4,096 | 5,120 | 6,144 | 7,168 | 8,192 |
| 64 | 4,096 | 8,192 | 12,288 | 16,384 | 20,480 | 24,576 | 28,672 | 32,768 |
| 128 | 16,384 | 32,768 | 49,152 | 65,536 | 81,920 | 98,304 | 114,688 | 131,072 |
| 256 | 65,536 | 131,072 | 196,608 | 262,144 | 327,680 | 393,216 | 458,752 | 524,288 |
| 512 | 262,144 | 524,288 | 786,432 | 1,048,576 | 1,310,720 | 1,572,864 | 1,835,008 | 2,097,152 |
| 1024 | 1,048,576 | 2,097,152 | 3,145,728 | 4,194,304 | 5,242,880 | 6,291,456 | 7,340,032 | 8,388,608 |
| 2048 | 4,194,304 | 8,388,608 | 12,582,912 | 16,777,216 | 20,971,520 | 25,165,824 | 29,369,128 | 33,554,432 |
| 4096 | 16,777,216 | 33,554,432 | 50,331,648 | 67,108,864 | 83,886,080 | 100,663,296 | 117,440,512 | 134,217,728 |
| 8192 | 67,108,864 | 134,217,728 | 201,326,592 | 268,435,456 | 335,544,320 | 402,653,184 | 469,762,048 | 536,870,912 |

# Spatial Resolution 1

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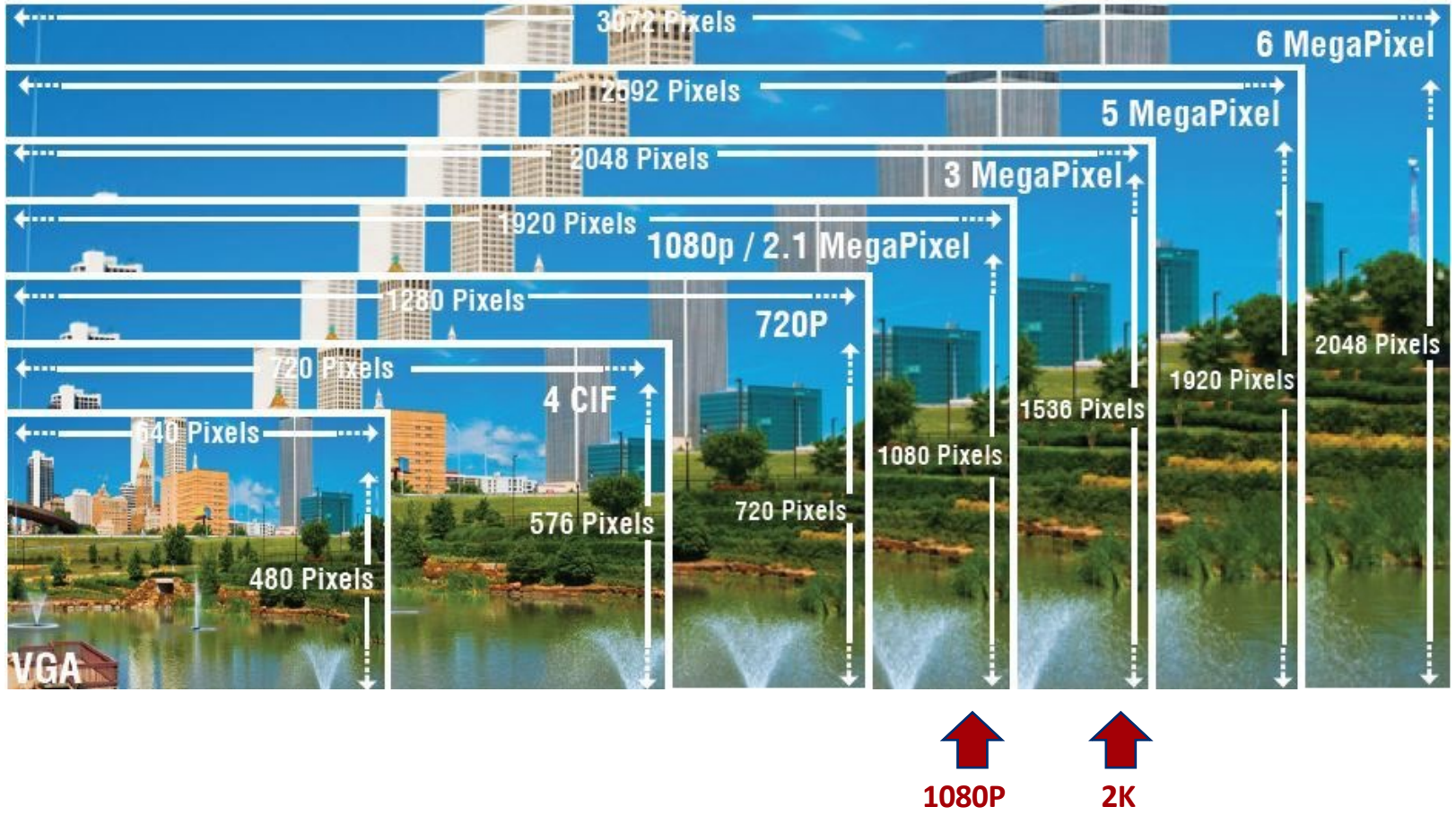
## Spatial Resolution: smallest discernible detail in an image



# Spatial Resolution 2

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## Spatial resolution – Sampling vs Size



# Example for Spatial Resolution

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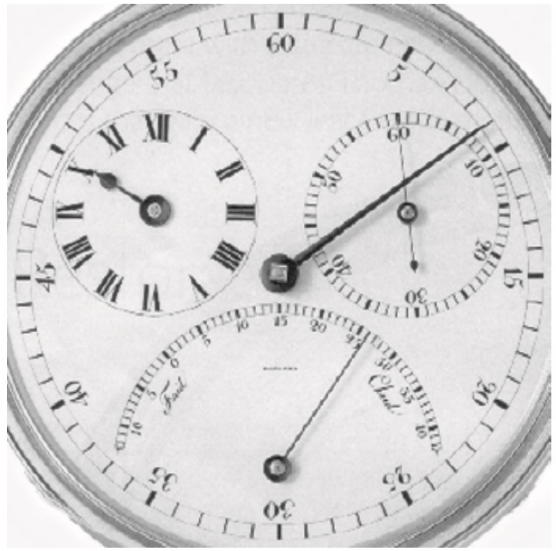
## Spatial resolution:  dpi (Dots Per Inch)



930 dpi



300 dpi

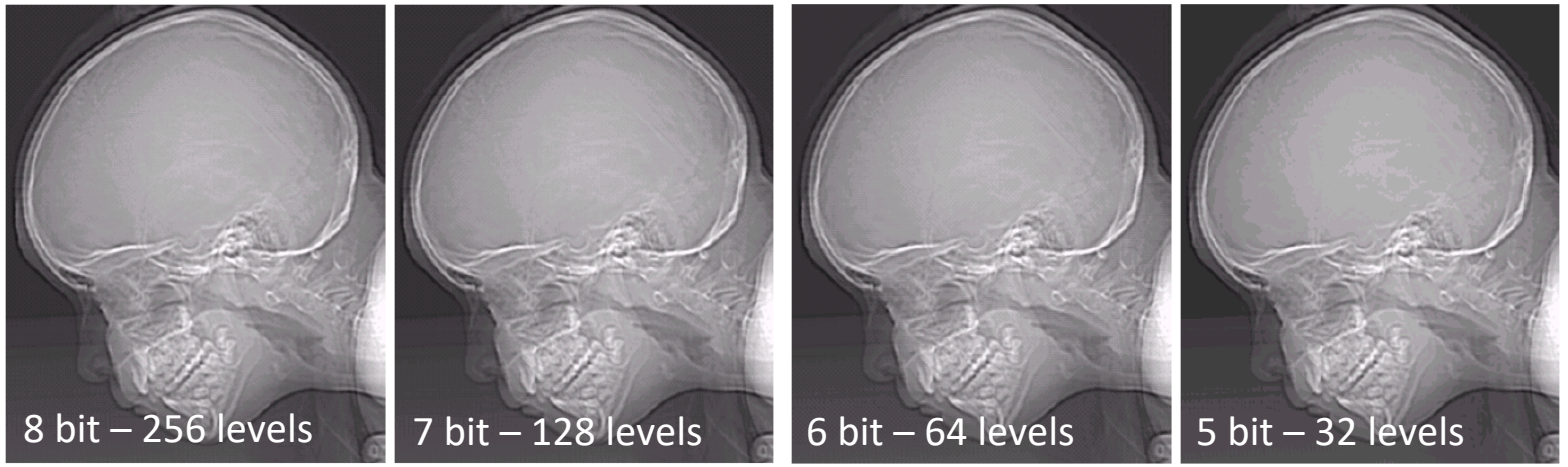


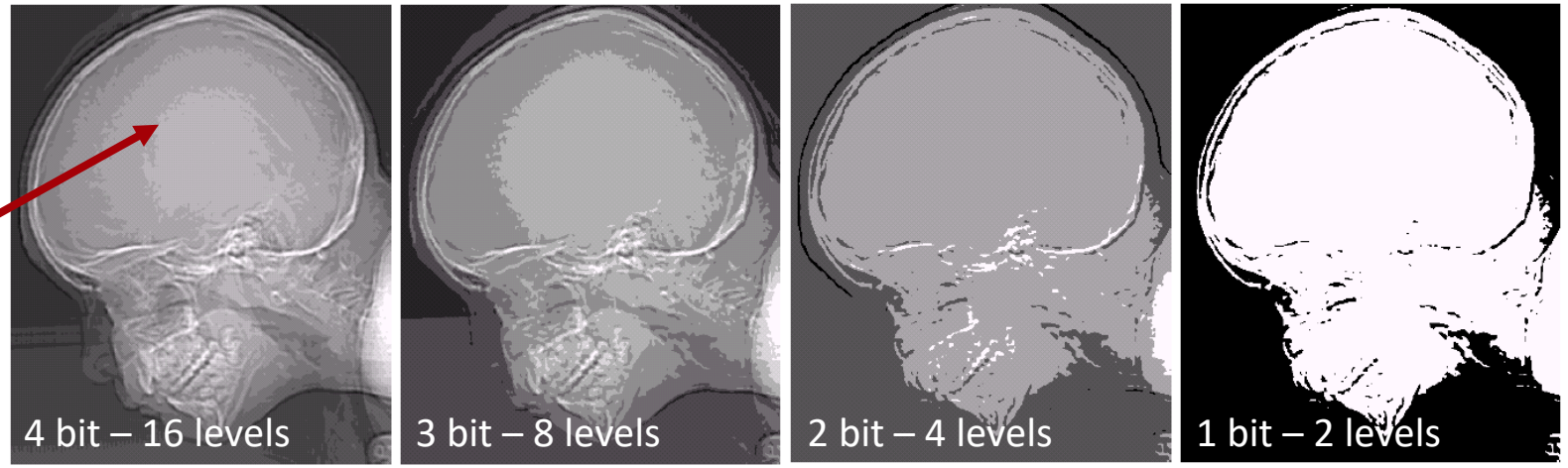
150 dpi



72 dpi

# Intensity resolution: smallest discernible change in intensity level




False contouring

# Further discussion of resolution

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- Imaging system resolution:
  - e.g. For optical imaging, optical resolution describes the ability of an imaging system to resolve details in the object that is being imaged. – wiki
  - Rayleigh criterion:

$$r = \frac{1.22\lambda}{2n\sin\theta} = \frac{0.61\lambda}{\mathrm{NA}}$$

- Temporal resolution: imaging speed



# What is light?

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- “Nature and Nature's laws lay hid in night:
  God said, Let Newton be! and all was light.” — Alexander Pope





Maxwell’s equations

$$\nabla \cdot \vec{D} = \rho_{\text{free}}$$

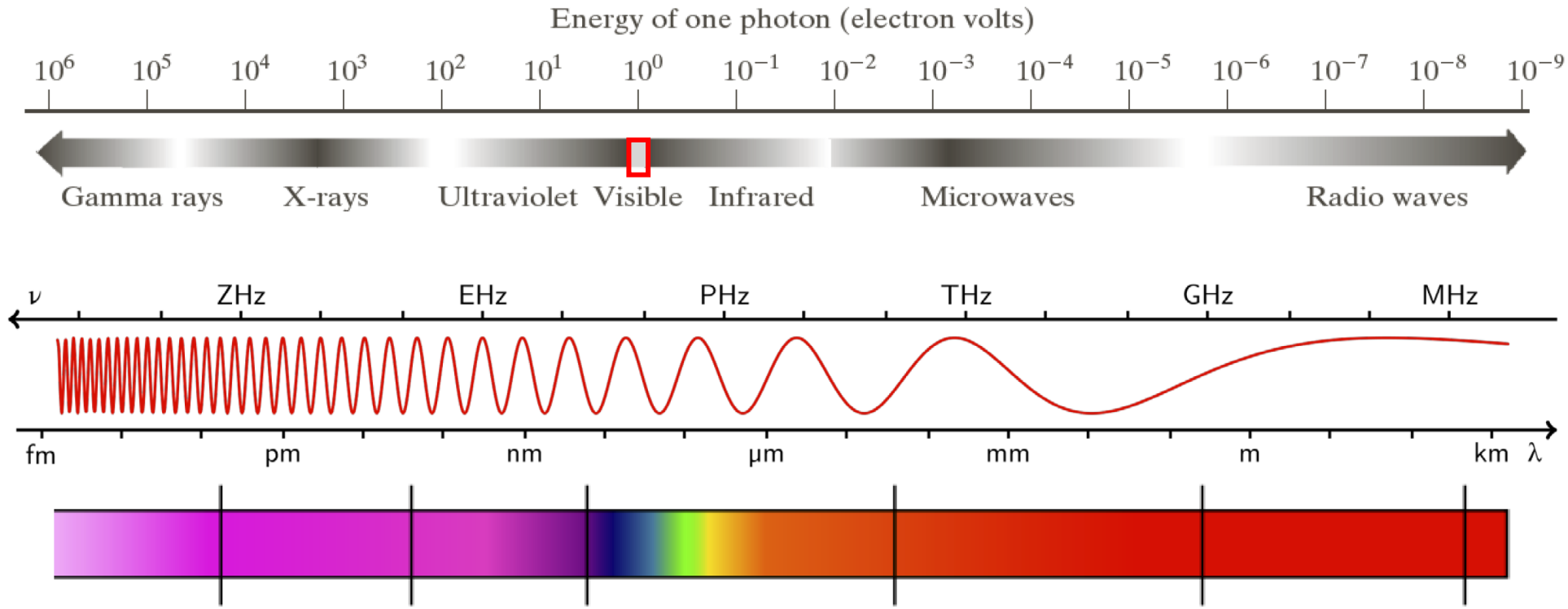
$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J}_{\text{free}} + \frac{\partial \vec{D}}{\partial t}$$

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# Electromagnetic spectrum

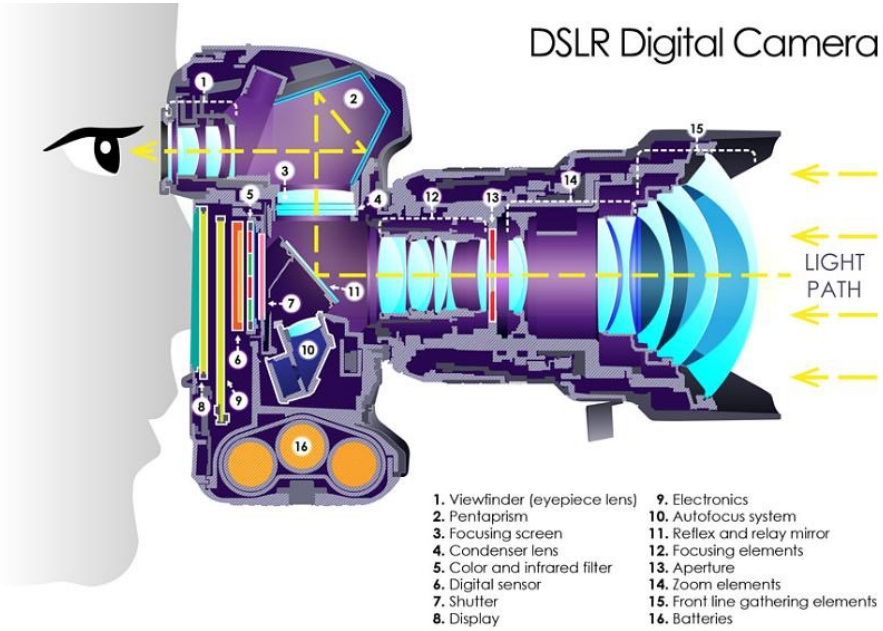





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# Visible-band (1 eV)







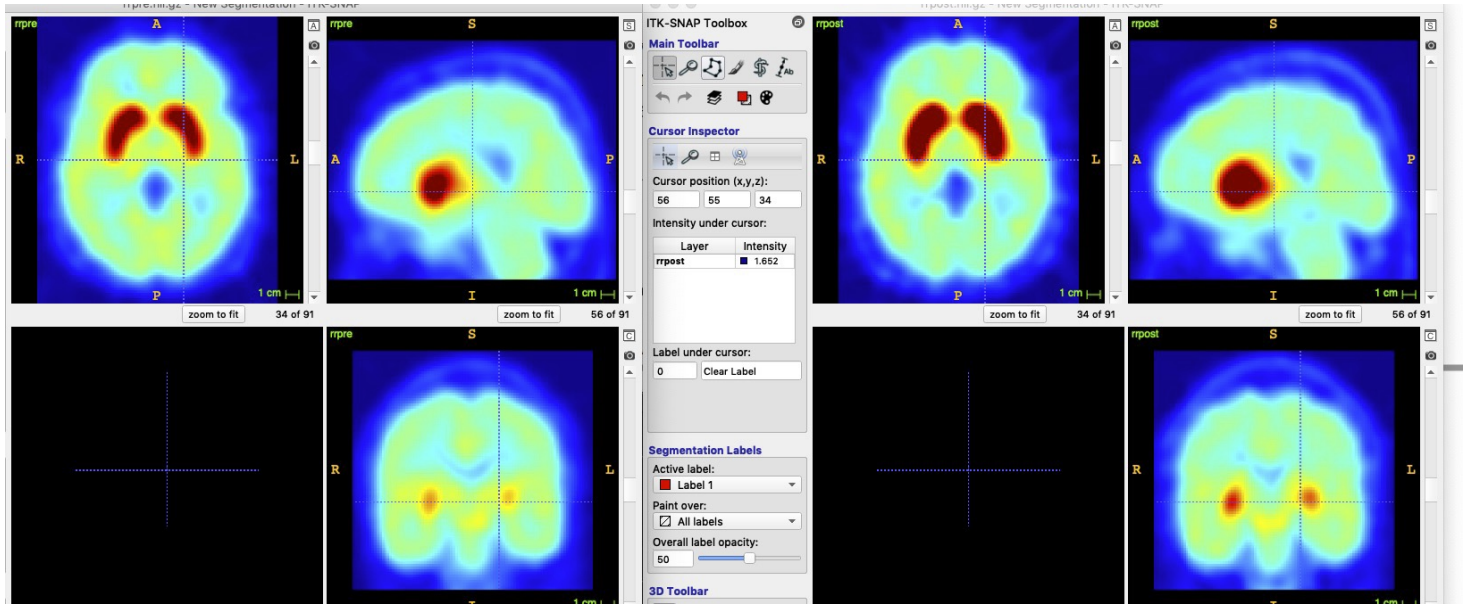




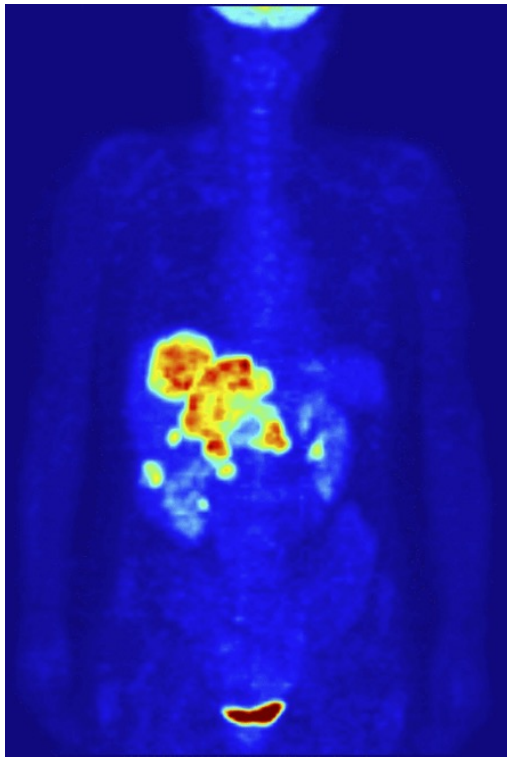
- Smartphones.
- Light microscope.
- Remote sensing (satellite).

# Gamma ray (105-106 eV)






Brain PET（Positron Emission Tomography） imaging from a drug addicted patient before and after DBS surgery.
Wei. et al. JNS 2019

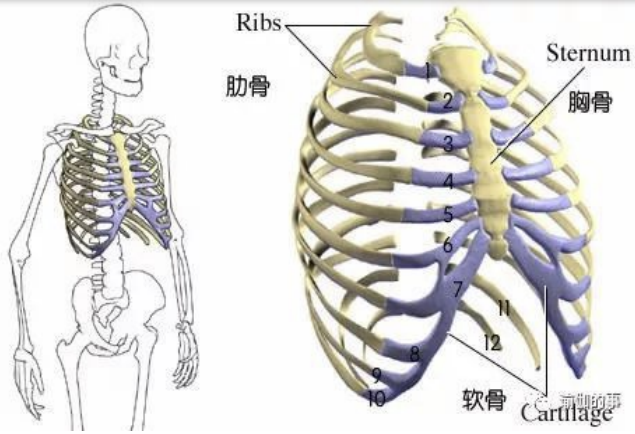


Whole body PET imaging with kidney labeled using 18F-FDG

# X-Ray (103-104 eV)





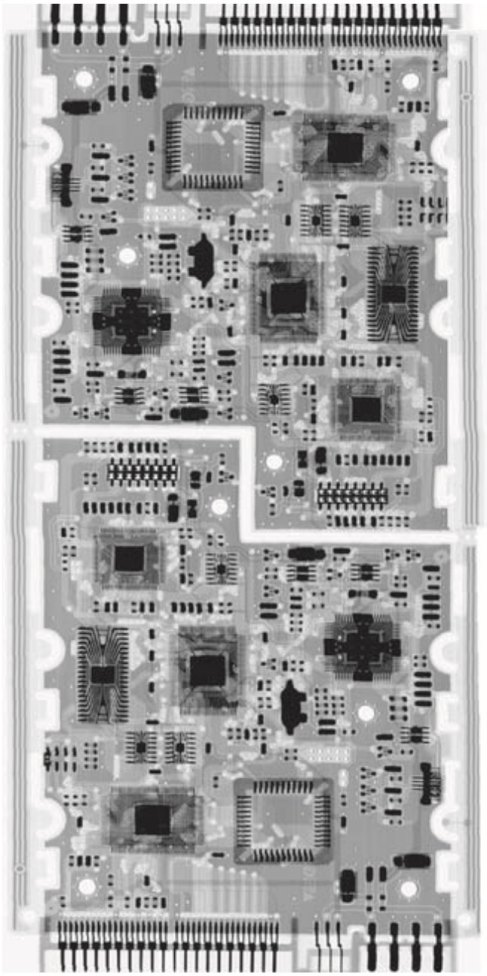





X-ray chest image




X-ray chest image of a Covid-19 Patient



Circuit boards

# X-Ray CT (103-104 eV)

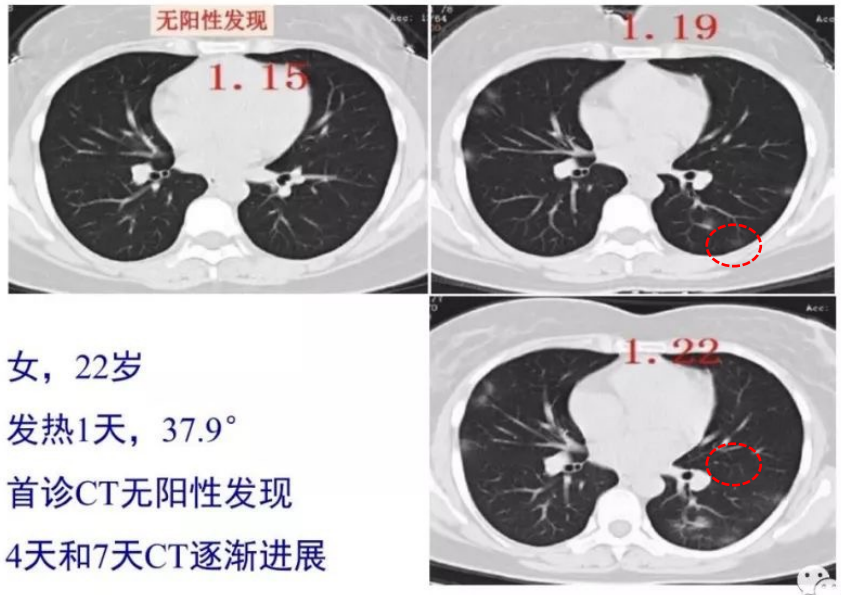


**CT(Computed Tomography)，**
**即电子计算式断层扫描**



早期CT表现-不特异

华中科技大学同济医学院 武汉市中心医院 THE CENTRAL HOSPITAL OF WUHAN




女，22岁

发热1天，37.9°

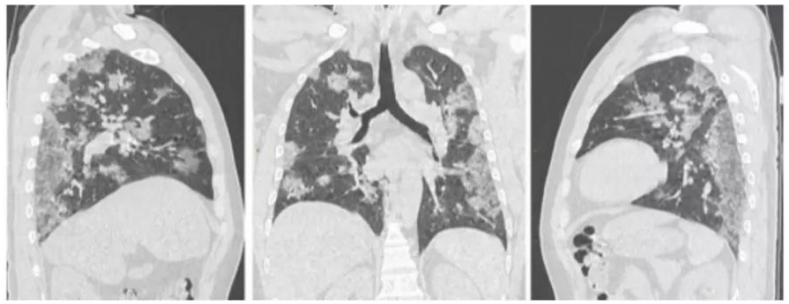
首诊CT无阳性发现

4天和7天CT逐渐进展

高少影像

早期CT表现

武汉协和放射 DEPARTMENT OF RADIOLOGY WUHAN UNION HOSPITAL



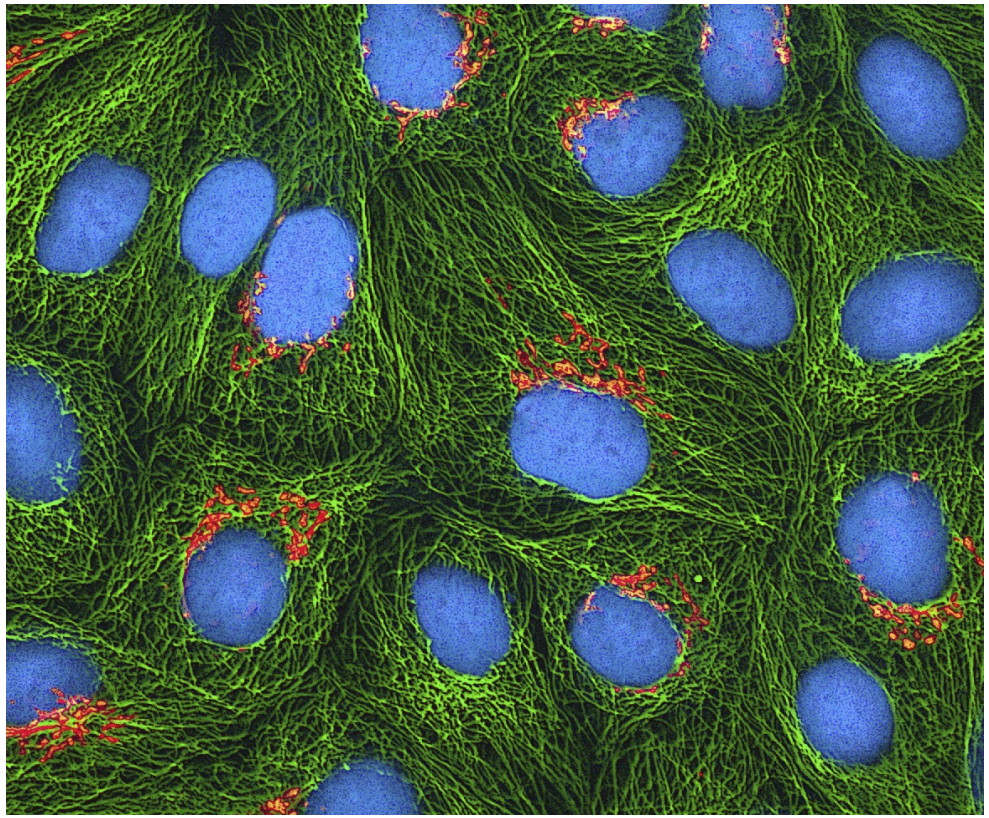
男，44岁，华南海鲜市场密切接触史

无明显诱因发热、乏力，外院抗菌抗病毒治疗无效

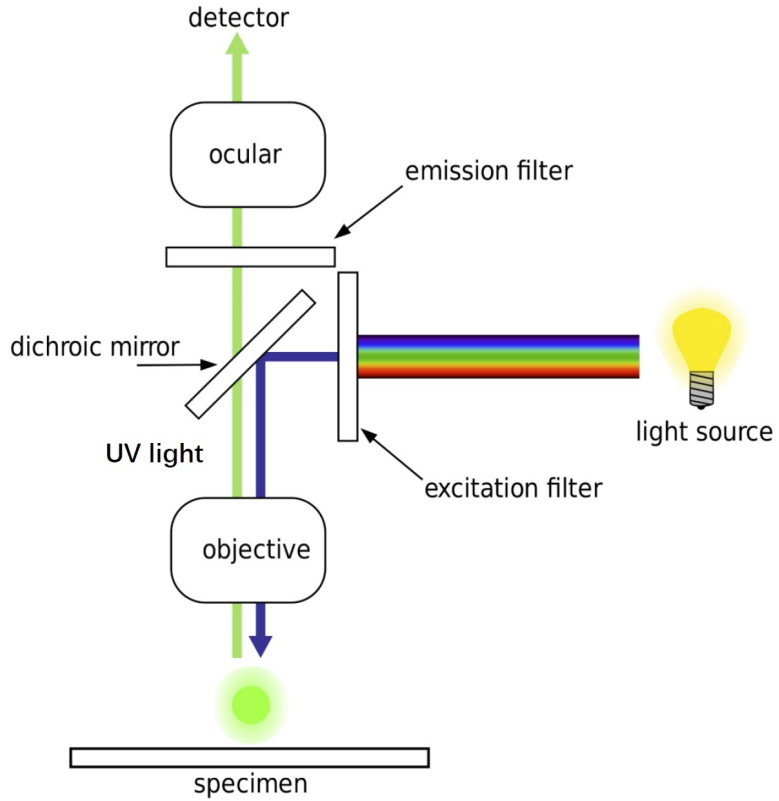
高少影像

# Ultraviolet imaging (10-100 eV)





Fluorescence microscopy image

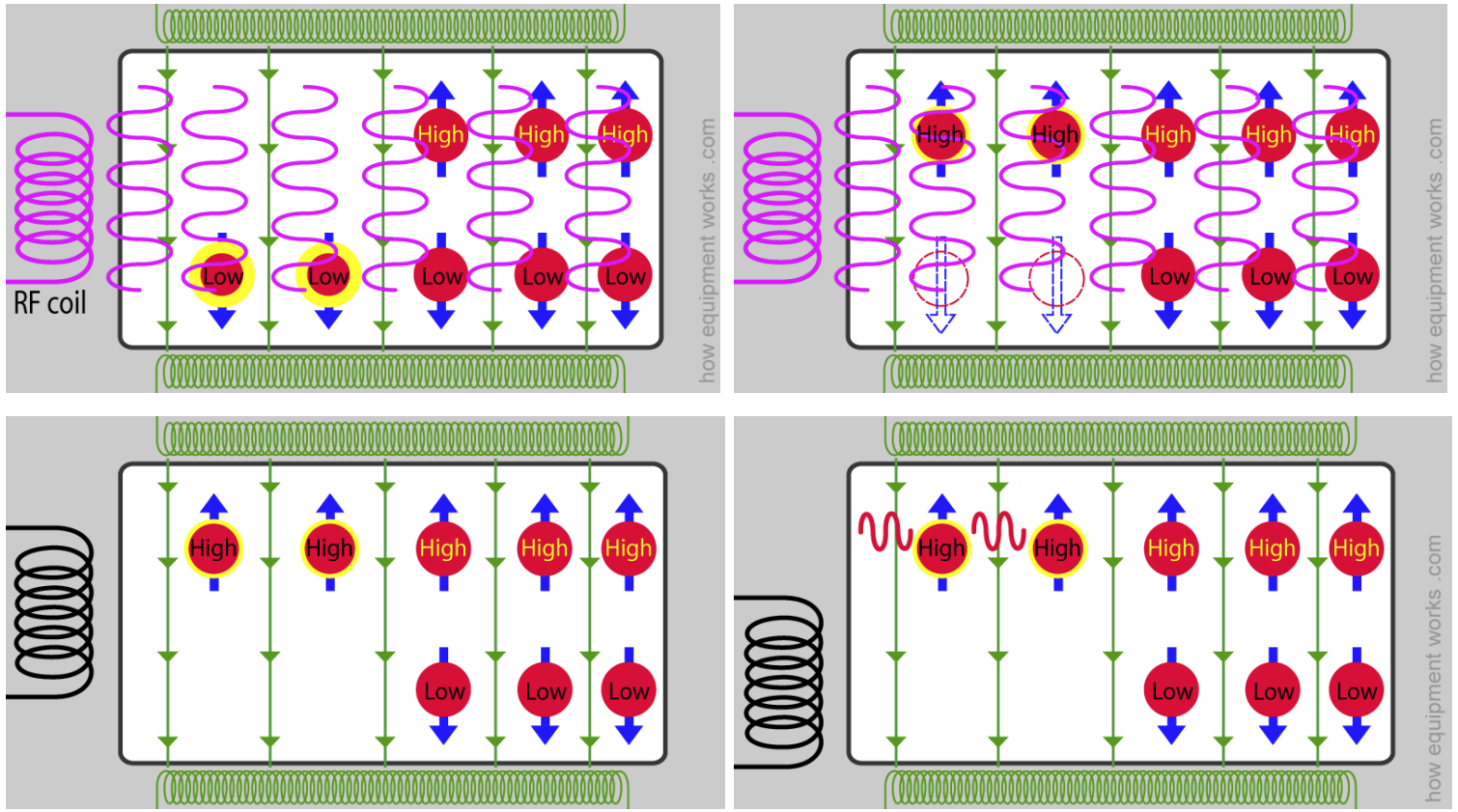


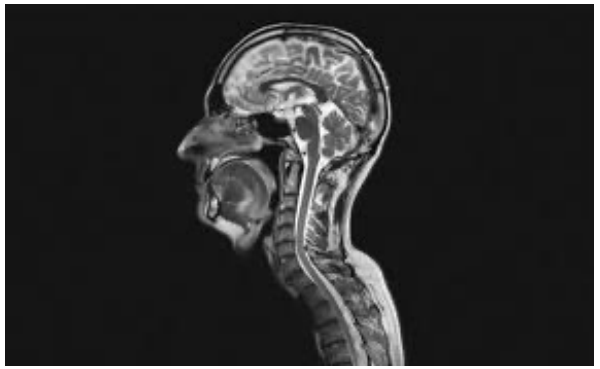

Fluorescence microscopy structure

# Magnetic resonance imaging (Radiofrequency/RF coil)

上海科技大学
ShanghaiTech University

## Radio-band imaging ($10^{-6}$-$10^{-9}$ eV)







立志成才报国裕民

# Ultrasound imaging (non-photon imaging)



